Course CS5.401

User Interaction and Usability of Digital Products

The WiseWanderer App

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Project Journey Document

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1 Introduction

Traveling alone can be one of the most powerful and life-changing experiences a person can have—particularly for women. It is a great opportunity to see the world on your own terms, learn who you are, and gain confidence. But alongg with the freedom comes the real issues too. Most women are concerned about their safety—what if they become lost, harassed, or in need of immediate assistance and don't know where to go?

While there are many apps to book hotels, view maps, or map out routes, few really address what independent travelers need most: the sense of security, assistance, and authentic human interaction. Some apps for safety are so cumbersome to employ when things go wrong. Others fail to give the type of real-time assistance or reliable information that travelers truly need. And when you're in an unfamiliar environment, it's difficult to know who—or what—you can count on.

That's where our steps in. We're building UI of a mobile app specifically for solo travelers, with women's safety at its core. It has features such as live location sharing with trusted friends and family, instant access to emergency assistance, connections to verified local guides, and the ability to leave genuine feedback about destinations you've traveled to.

2 Our HMW Statement

How might we empower solo travellers, especially women, to feel more secure by providing tailored safety solutions? Our How Might We statement addresses a central problem: how might we make solo travelers—particularly women—feel more supported and safer on the road? Solo travel is great option, but it's also filled with safety issues that aren't all being met by current tools. With this question, we're trying to answer and design solutions to safety that are not only useful, but also personal, trustworthy, and simple to use. It's about providing travelers with peace of mind, confidence, and a feeling of belonging—even when they're traveling alone.

3 Stakeholders

Primary Stakeholders: These are the direct users of the product. They actively use the app and are the most affected by its features and usability.

- Solo Travelers (especially women)
- Tourists and Backpackers
- LGBTQ travellers
- Domestic and International Travelers
- Travel Influencers / Content Creators
- Business Travelers

Secondary Stakeholders: These are indirect users or those who support or are connected to the experience of the primary users.

- Trusted Contacts / Family & Friends
- Verified Local Guides
- Emergency Services / Police
- Travel Agencies
- UX Researchers / Designers
- Product Development & Support Teams

Other Stakeholders: These are external groups or entities who have a broader interest or influence in the product's outcomes.

- Government Tourism Departments
- NGOs / Safety Advocacy Groups
- Data Privacy Consultants
- Transport Services / Ride-Hailing Apps
- Accommodation Providers (Hostels, Hotels)
- Technology Providers (APIs, Alerts)
- Investors / Sponsors
- Regulatory Bodies

4 Literature Review Summary

As part of our study, we reviewed 15 papers on women's safety, solo travel, and the importance of UI/UX in addressing these concerns. We discovered that for women, travel safety isn't only about staying away from harm—it's also about being confident, assisted, and mentally comfortable during travel.
Throughout the articles, there were certain safety issues that repeatedly surfaced. Most women are concerned about physical threats such as harassment, theft, or assault. Public transport may not be safe, and it is usually difficult to find safe accommodations. In certain places, there is inadequate infrastructure—poorly lit roads, no emergency assistance, and few safe ways to get around. Besides physical danger, psychological barriers also complicate travel. Most women are frightened or nervous when they are alone somewhere new. Some are afraid of being judged for going someplace by themselves, particularly somewhere it's not very usual. They're also afraid of being lonely, as it may be difficult to meet people they can trust. Online dangers were also discussed. Some women are harassed online while they are traveling—like strange messages or apps that follow them. In addition to that, it is hard to believe online reviews or details about how safe a place is because much of it might be false or ambiguous. And, to make themselves feel safer, women end up spending extra money—for instance, taking private transport rather than public, or staying at pricey hotels with good security. They might not know where to seek emergency assistance, legal aid, or women-friendly services in foreign locations.

4.1 What Can Be Done to Improve the Situation?

Researchers identify some helpful measures that can make it easier and safer for women to travel alone. Improving infrastructure is the first step. Women-only transport, safer hostels with female-only rooms, and emergency help desks can make a huge difference. Businesses and local governments need to prioritize installing security features such as 24/7 surveillance, improved locks, and transparent emergency procedures. Technology also has an important role to play. Apps can provide functionality such as AI-driven route planning, panic buttons, and sharing location. GPS-enabled wearable devices and alert mechanisms can provide an additional level of safety. It's also worth providing travelers with access to real, candid safety information. Finally, a strong community can make women feel less isolated. Travel sites can post cultural advice, dress codes, and safety information for various areas. Having safe local guides or a buddy system can provide feelings of protection and alleviate fear.

4.2 Role of UI/UX in Supporting Solo Women Travelers

Good design can enhance these safety solutions to be more effective. For instance, maps with safety scores for neighborhood areas can assist travelers in selecting safer routes. Apps can prevent mazes of confusing directions and instead provide straightforward, easy-to-understand navigation—particularly in emergencies. Elements such as one-tap SOS buttons, location-based alerts, and emergency contact sharing can be incorporated into the user interface directly. A simple and intuitive design makes users find key features easily when necessary. UI/UX may also enable community-building by providing a secure environment in which to engage with fellow travelers or local guides who have been vetted. In general, careful design may enable women to feel more confident and in command while traveling alone.

5 User Research

User research is the process of finding out about the individuals who will be using our product. It enables us to know about their needs, issues, behavior, and interests so that we can create something that actually works for them. This may involve techniques such as surveys, interviews, observations, or usability testing(at later stage).

5.1 Research objective

The primary research objective of this study is to comprehend the safety issues of solo travelers, particularly women, and investigate how a mobile application with intelligent and caring UI/UX elements can make them feel safer, more confident, and connected while traveling. This involves determining critical issues, researching current solutions, and creating a product that offers real-time assistance, reliable information, and a sense of belonging.

5.2 User Research: Phase 1

We conducted a survey to comprehend the needs, concerns, and habits of solo women travelers. A survey was an ideal option for Phase 1 of this project because it helped us reach a large number of solo women travelers in a short and effective span of time. Surveys also gave us quantitative insights, enabling us to find common pain points, needs, and preferences among users. This allowed us to establish a solid foundation for additional investigation and design choices. The survey consisted of the following questions:

1. "What is your age group?"
2. "What is your gender?"
3. "How often do you travel alone?"
4. "What are your primary reasons for solo travel? (Select all that apply)"
5. "Which types of destinations do you usually visit?"
6. "What does the word 'safe' mean to you when you are travelling solo or in a group too?"
7. "Have you ever felt unsafe while traveling alone?"
8. "What safety concerns do you face while traveling alone? (Select all that apply)"
9. "What safety measures do you currently take while traveling solo? (Select all that apply)"
10. "Do you currently use any digital safety tools while traveling?"

11. “What features do you find most useful in a travel safety app?”
12. “How important is community support for your solo travel safety?”
13. “Would you trust real-time, user-generated safety ratings for locations?”
14. “Would you use a platform that connects solo travelers with verified local guides or buddy travelers?”
15. “What features would you like to see in an ideal travel safety platform?”
16. “How would you prefer to receive safety alerts while traveling?”
17. “Would you be willing to pay for a premium version of a travel safety app with advanced security features?”
18. “What improvements do you think should be made in existing travel safety solutions?”

5.2.1 Pain Points

From the survey responses, we identified several key pain points faced by solo women travelers:

- **Safety Concerns:** Respondents worry about theft, scams, physical assault, and unsafe transport.
- **Negative Experiences:** Many women have reported feeling unsafe during their travels.
- **Current Solution Limitations:** Respondents feel that existing safety apps lack strong verification systems, making them less reliable.
- **Tool Adoption Issues:** Some women don’t use digital safety tools at all, which indicates a gap in awareness or trust.
- **Complex Interfaces:** Some digital safety tools are overly complicated, causing hesitation in full reliance on them.

5.2.2 User Needs

The survey also highlighted the following key needs for solo women travelers:

- **Reliable Alerts:** Travelers need real-time safety alerts, including features like emergency SOS.
- **User-Friendly Tools:** Digital tools need to be easy to use, with minimal complexity.
- **Proper Verification:** Travelers require verified and trustworthy safety information.
- **Effective Communication:** Users need clear and timely methods to receive safety alerts.
- **Community Feedback:** A system that supports verified user-generated safety ratings and fosters community support is essential.

5.2.3 Key Patterns

From the responses, we observed the following key patterns:

- **Demographics:** Most respondents fall within the 18-25 and 40+ age groups.
- **Travel Frequency & Destinations:** Many solo travelers travel around 2-4 times a year, with a few traveling rarely.
- **Destination Preferences:** Respondents prefer urban destinations, often mixed with domestic travel.
- **Primary Motivation:** Leisure or vacation is the most common reason for traveling solo.
- **Important Features:** Real-time location sharing and emergency SOS were the top requested features.
- **Community Importance:** Community support and trust in user-generated safety ratings are highly valued when exploring new locations.

5.3 User Research: Phase 2

In Phase 2, we did interviews to better understand the needs, concerns, and preferences of individual women travelers. Interviews were well-suited for Phase 2 because they enabled more individualized, qualitative findings. By speaking directly with travelers, we could identify subtle pain points, motivations, and specific opinions, which were useful for further honing our design. The interviews were structured into a number of sections to address important elements of the solo travel experience, safety issues, and the use of digital tools. The interview questions were:

1. "What age group do you belong to?"
2. "How frequently do you travel alone, and what factors influence this?"
3. "What motivates you to travel solo?"
4. "Can you describe the types of destinations you prefer to visit and why?"
5. "Can you share an experience where you felt unsafe while traveling alone?"
6. "What are the most common safety concerns you consider before traveling alone?"
7. "How do you assess the safety of accommodations and transport before choosing them?"
8. "How do digital tools influence your sense of safety while traveling alone?"
9. "What digital tools or safety apps do you currently use while traveling?"
10. "What features do you currently find most useful in travel safety apps, and what additional features would you like to see?"
11. "How do you evaluate the reliability of safety recommendations from AI or digital platforms?"
12. "Would real-time, crowdsourced safety ratings influence your travel decisions? Why or why not?"
13. "How do online communities and peer networks influence your travel safety decisions?"
14. "How would having access to verified local guides impact your sense of safety and travel confidence?"
15. "What are your thoughts on buddy systems for solo travelers? Have you used one before?"
16. "If you could design an ideal solo travel safety tool, what features would it include?"
17. "Based on your experiences, what improvements would you suggest for digital tools designed to enhance solo travel safety?"
18. "What kind of support or solutions would help you feel more secure when traveling alone?"

5.3.1 Pain Points

From the interview responses, we identified several pain points faced by solo women travelers:

- **Unreliable Safety Indicators:** There were gaps in real-time safety data, such as drivers taking unfamiliar routes, delayed assistance from apps, and inconsistent user-generated reviews.
- **Lack of Information from Digital Tools:** Many digital tools did not meet all user needs, leading to confusion and ineffective solutions.
- **Varied Alert Preferences:** Some users preferred SMS alerts due to network issues, while others preferred in-app notifications, creating uncertainty about the most effective alert methods during emergencies.
- **Skepticism Towards AI:** Users showed distrust towards AI-driven safety recommendations, preferring verified reviews and human-generated feedback.

5.3.2 User Needs

The interviews highlighted the following key needs for solo women travelers:

- **Trusted and Verified Feedback:** Platforms need to offer accurate, live safety updates, crowdsourced ratings, and verified comments.
- **Real-Time Alerts:** Alerts should function even in low-connectivity scenarios.
- **Integrated, User-Friendly Tools:** Travelers expect seamless safety solutions that combine features like live location sharing, emergency SOS, and easy navigation in a single app.
- **Trusted Community Input:** Users need platforms that feature authentic feedback from fellow travelers, ensuring that safety ratings and recommendations are trustworthy.
- **Premium Feature Value:** Users are willing to pay for premium features if they offer key tools like offline support, direct emergency service integration, and verified local authorities access at an affordable price.

5.3.3 Key Patterns

From the interviews, the following key patterns emerged:

- **Urban Travel:** Most travelers prefer urban destinations, often combined with domestic travel.
- **Dependence on Reviews & Peer Feedback:** Interviewees heavily rely on reviews, ratings, and recommendations, emphasizing the importance of community-generated insights.
- **Diverse Digital Tool Usage:** Frequent solo travelers use a variety of digital tools, such as transaction tools, location sharing, and ride-hailing apps. In contrast, occasional travelers rely more on personal judgment and selected trusted platforms.
- **Skeptical Towards AI:** Interviewees preferred human-verified feedback over AI-driven recommendations, as they believe AI should not replace personal experiences.

Overall, the combined findings of Phase 1 and Phase 2 exposed key trends in the solo female travelers' behavior. The Phase 1 survey gave general demographic and behavioral information, emphasizing real-time notifications, user ratings, and a high demand for uncomplicated, user-friendly digital safety solutions. The survey also discovered key pain points, including irregular safety information and the absence of a single, trustworthy safety solution. Phase 2, through in-depth interviews, provided richer personal experiences and particular challenges. They explained how they use safety apps now, what they like about receiving alerts, and stressed that they trust human-vetted feedback over AI-powered suggestions. Both phases conveyed the necessity of comprehensive, dependable, and convenient safety measures to make solo travelers feel safe and confident during their trips.

Category	**Phase 1: Survey (50 Responses) – Broad Insights**	**Phase 2: Interviews (6 Users) – Deep Insights**
Safety Concerns	Fear of harassment, unsafe transport, theft, scams.	More detailed concerns like **stalking, unwanted attention, and digital safety threats.**
Use of Safety Tools	Many travelers are **not using safety apps** regularly due to lack of awareness.	Users who do use safety tools often **find them ineffective or lacking real-time support.**
Emergency Response	Slow SOS responses and limited real-time safety features.	More **nuanced frustrations with emergency services,** like **delayed police action or lack of local support.**
Accommodation Safety	General distrust in host verification processes.	More **specific concerns about women's safety in mixed-gender accommodations.**
AI & Tech Integration	Interest in AI-driven safety solutions.	**Skepticism about AI-driven recommendations;** prefer a **mix of human verification and AI insights.**
Preferred Safety Features	Community-based solutions like **crowdsourced safety ratings.**	Preference for **verified local guides, women-only spaces, and direct government integrations.**

6 Insights & Trends

As all the interviewees were women, their answers had greater emotional intensity than the survey. Several of them provided firsthand accounts of harassment, unsafe housing, and transport-related anxiety, so personal safety was the most recurring concern. Participants resoundingly rejected excessive dependence on AI-driven recommendations, instead focusing on faster emergency response, human-checked safety networks, and direct liaison with authorities such as the police. They showed a preference for active, system-provided warnings and reliable safety networks as compared to sites depending exclusively on self-reporting. Drawing on the insights of both research phases, we determined four repeated traveler personas that remained relevant:

1. **The Minimalist, Budget-Value Traveler**, who desires key features at minimal expense
2. **The Tech-Savvy, Full-Feature User,** who is willing to embrace sophisticated features and innovation
3. **The Community & Information Driver**, who depends on peer feedback and aggregated safety information
4. The Cautious Traveler, who values vetted information and favors solutions with transparent policing or authority integration.

A User Journey Map (UJM) is a detailed visualization of a user's experience with a product, service, or system, broken down into different stages of interaction. It captures the user's actions, emotions, pain points, and touchpoints across the journey, helping to understand their needs, expectations, and frustrations at each step.
We have made a generalised UJM of a solo female traveller irrespective of her user persona. Persona wise UJM's can be seen in "Insights and Trends" document already submitted on moodle.

User Journey Map - Solo Female Traveler Safety

Stage	Awareness	Consideration	Planning & Booking	Travel & Navigation	On-Trip Experience	Post Experience
User Actions	Thinks about safety risks while planning travel.	Researches destinations, checks safety ratings & tips.	Books accommodation, transport, and safety tools.	Uses navigation apps, stays alert to surroundings.	Engages in activities while monitoring safety.	Shares experiences, safety feedback online.
Thoughts & Feelings	"Is this destination safe for solo travel?"	"I need reliable safety info before making decisions."	"I hope my accommodation and transport are secure."	"Am I being followed? Is this cab safe?"	"I want to enjoy, but I must stay cautious."	"Should I warn others about this place?"
Touch Points	Travel forums, social media, word-of-mouth.	Safety apps, government advisories, reviews.	Booking platforms, safety tool apps, GPS tracking.	Maps, ride-sharing apps, emergency contacts.	Local guides, buddy systems, safety check-ins.	Travel groups, safety rating platforms.
Pain Points	Lack of reliable, real-time safety information.	AI safety ratings feel generic, not tailored.	Booking sites lack verified security details.	Unreliable public transport & cab safety.	Safety concerns in public spaces, scams.	No follow-up from safety platforms.

7 Feature recommendations

Based on the literature review and both phases of user research, we propose a set of features designed to address the real-world challenges faced by solo women travelers. The survey responses emphasized the need for real-time alerts, user-friendly tools, and community-driven safety insights. Interviews reinforced these findings with emotionally grounded accounts, highlighting the demand for human-verified networks, responsive emergency features, and trust-based platforms. The following seven features were selected and designed in direct response to these findings:

- **Sign-in & Onboarding:** A secure and intuitive onboarding process helps users quickly set up their profile, safety preferences, and trusted contacts. Given the need for simplicity and clarity in digital tools, this feature ensures better adoption and accessibility, especially for first-time users.
- **Trusted Contacts:** This feature allows users to share live location and emergency alerts with selected friends or family. Both survey and interview participants emphasized the importance of reliable direct communication channels to ensure peace of mind while traveling.
- **Post-Trip Feedback:** After completing a trip, users can share safety feedback about destinations, accommodations, and transportation. This feature builds a repository of verified, community-generated data, an aspect highly valued by users who distrust purely AI-generated recommendations.
- **Search & Explore:** Users can search for destinations using safety ratings, reviews, and tips from other travelers. The importance of crowd-sourced safety information and peer reviews was repeatedly mentioned, especially when planning travel to new or unknown locations.
- **Buddy System / Verified Local Guides:** This feature connects users with vetted locals or fellow solo travelers nearby. The insights of the interviews revealed a strong preference for human support and community engagement, and users expressed more confidence in networks involving verified individuals.
- **Emergency Assistance & SOS:** Users can trigger an SOS alert, notify emergency contacts, and access local help. Emergency response capabilities, especially those that work offline or with weak connectivity, were highlighted as essential for travelers' safety.
- **Real-Time Navigation with Safety Alerts:** This feature provides live navigation support with integrated safety alerts, such as high-risk areas or user-reported incidents. Users preferred proactive, context-aware guidance over generic self-reporting tools, especially in unfamiliar environments.

These characteristics are intended to develop a cohesive and responsive safety platform that incorporates usability, trust, and community feedback. They are not only intended to inform, but also to empower individual travelers with real-time, credible, and actionable safety assistance.

7.1 Approach: Jobs-To-Be-Done (JTBD) Framework

For our "How Might We" (HMW) statement and subsequent feature recommendations, we have adopted the Jobs-To-Be-Done (JTBD) framework. This approach offers a more user-centric lens by focusing on the underlying motivations behind users' needs and actions. Rather than simply addressing surface-level complaints, JTBD allows us to design solutions that fulfill real jobs solo travelers are trying to get done.

- **A More User-Centric Approach:** JTBD prioritizes the *why* behind user needs. Solo travelers are not merely looking to avoid risks—they seek the freedom to explore independently with confidence. This framework shifts the design focus from reactive fixes to proactive support.
- **Alignment with Real-World Travel Scenarios:** The JTBD method reflects how users navigate through actual travel situations. For example, a solo traveler's job might not just be to "avoid unsafe areas" but rather to "explore new places with peace of mind." This helps us shape solutions that support the broader context of travel.
- **Safety as a Continuous Journey:** Safety is not a one-time event, but a continuous need: from planning and decision making to navigation and emergency response. JTBD enables mapping the entire traveler journey to identify where safety interventions are most critical and effective.

This structured approach ensures that our feature recommendations are directly tied to core user jobs, pain points, and expected outcomes. Each proposed feature, whether it is real-time navigation, emergency SOS, or verified guide connections, serves a specific user objective rooted in emotional and practical needs.
Ultimately, applying the JTBD framework helps us design a user-aligned seamless safety experience that goes beyond just solving problems. It builds trust, encourages community engagement, and empowers solo travelers—especially women—with reliable tools, support networks, and peace of mind. The result is a more effective, adopted, and impactful solution in the realm of digital travel safety.

8 Solution

Most Dominant Pain Points & Needs :

1. **Safety Anxiety**: Constant fear of theft, scams, and personal harm, especially when navigating unknown or dimly lit areas.
2. **Distrust in Current Tools**: Existing apps are either too complex, lack transparency, or fail to provide verified, relevant data.
3. **Fragmentation of Features**: Users expressed frustration about needing multiple apps for booking, navigation, reviews, and emergency support.
4. **Desire for Human Validation**: Community-sourced feedback and recommendations were seen as significantly more trustworthy than algorithm-based suggestions.
5. **Emergency Readiness**: A clear gap exists in accessible and immediate SOS features during moments of panic.

8.1 Problem Statement and Connection to HMW

Solo travellers, especially women, face persistent challenges related to safety, trust, and navigation. The current ecosystem of travel safety apps is often fragmented, unverified, or overly complex—leaving users overwhelmed and underprotected.
Our proposed solution directly addresses the HMW (How Might We) statement:

> *"How might we empower solo travellers, especially women, to feel more secure by providing tailored safety solutions?"*

This solution aims to bridge critical gaps in the travel safety landscape by offering a digital platform that:

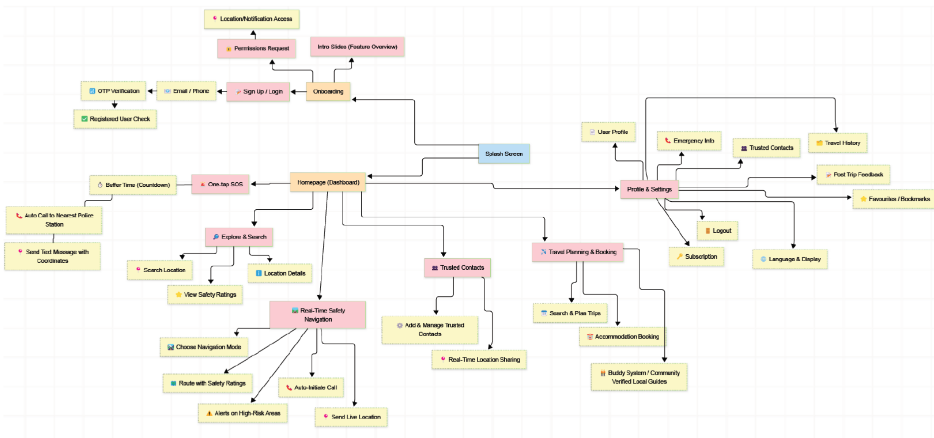
- Merges real-time navigation, safety alerts, and community insights in one place.
- Provides verified, human-centric data rather than unfiltered or AI-generated suggestions.
- Functions seamlessly in both online and offline environments to ensure reliability at all times.
- Maintains a clear, approachable, and customizable interface for users of varying tech-savviness.

Our intended mobile app serves as a single, safety-oriented travel assistant adapted for single travelers. It consolidates key features like trip planning, peer feedback, real-time routing, emergency contact, and trusted contacts into a single, cohesive system. Designed through the lens of extensive user research learnings and rooted in usability principles, the app keeps the traveller smoothly connected to important features at every stage of travel. It provides real-time responses in both networked and low-connectivity environments, encourages social interaction through confirmed reviews and buddy systems, and gives users a sense of personal empowerment through providing them with increased control, consciousness, and ongoing assistance during travel experiences.

9 Design

9.1 Information Architecture

Following is the information architecture of our app- meaning how will the information flow within the app and how will the users can navigate within the app and its pages.



Our Homepage (Dashboard) acts as the central hub, from which users can branch out to: Each of these

Module	Purpose
Travel Planning	Book safer travel and accommodation, access verified information.
Explore/Search	Discover destinations with crowdsourced safety scores.
Safety Navigation	Real-time route tracking with hazard alerts.
Trusted Contacts	Share live location, auto-alert in emergencies.
Profile & Emergency Info	Set up personal information, access post-trip logs, manage subscriptions.

Table 1: Key Modules and Their Purpose

features can be accessed independently and returns the user to the central hub, keeping the experience anchored and intuitive.

9.1.1 Use Cases

Based on our literature review, user surveys, and in-depth interviews, we identified the following key use cases for solo travelers—especially women—where safety, trust, and ease of use are critical:

1. **Real-Time Navigation with Safety Alerts**
2. **Emergency Assistance and SOS Trigger**
3. **Post-Trip Feedback and Incident Reporting**
4. **Trusted Contacts with Live Sharing**
5. **Safe Travel and Accommodation Planning**
6. **Buddy System for Local Guidance and Group Travel Options**

9.1.2 Implementation Framework: Jobs-To-Be-Done (JTBD)

The following table outlines how our key use cases align with the JTBD framework by mapping the core user jobs with their associated pains, desired gains, and proposed solutions:

Job To Be Done	Pain	Gain	Solution
Plan a solo trip safely	Lack of verified data, scams	Confidence in bookings	Community-verified booking system
Navigate a new place	Unsafe areas, fear of getting lost	Real-time support	Safety alerts + route guidance
Get emergency help	Delay in response, panic	Quick, reliable help	SOS button, auto-call, SMS backup
Choose safe places	Inconsistent reviews	Trustworthy info	Crowdsourced and verified reviews
Stay connected	Fear of being alone	Peace of mind	Trusted contacts and live location sharing
Share experiences	Help future travelers	Collective safety	Post-trip feedback and ratings

Table 2: Use Case Alignment with JTBD Framework

9.1.3 Navigation Model: Hub-and-Spoke

The proposed mobile application follows a Hub-and-Spoke navigation model, where a central hub (typically the home screen or dashboard) connects users to multiple independent modules such as Travel Planning, Explore, Navigation, Trusted Contacts, and Profile. This model is particularly well-suited for safety-critical apps, as it provides a predictable and clear structure that minimizes user confusion, especially during emergencies. Each feature can be accessed directly from the hub without unnecessary depth or complexity, enabling quick transitions and reducing cognitive load. In the context of solo travel, where fast access to emergency features, trusted contact sharing, and verified information is essential, the Hub-and-Spoke model ensures that users are always just one tap away from critical functionality. Its simplicity and reliability make it a perfect navigation framework for building user trust, enhancing usability, and ensuring a secure travel experience.

9.2 Wireframes

Wireframes are a fundamental part of the design process, providing a clear, low-fidelity representation of the app's layout and structure. Created in Figma, these wireframes focus on the functional aspects of the app, illustrating how key features are placed within the interface. The goal was to prioritize user flow and interaction rather than visual design elements such as color, typography, or imagery.
Wireframes act as the "blueprint" of our application, which allowed us to focus on core usability and feature placement before diving into detailed design. They served as a quick way to validate key interactions and ensure that the layout aligned with user needs. We specifically used wireframes to solve pain points identified from user research, ensuring that the design directly addressed solo women travelers' safety concerns.

The design of the low-fidelity wireframes remains consistent throughout, ensuring a user-friendly and functional experience. Each feature has been carefully crafted based on user insights and feedback, with a focus on simplicity, accessibility, and usability. For instance, the Dashboard provides a central hub for accessing all essential features like emergency assistance and trip status. The Incident Reporting screen offers an easy-to-use form for sharing safety incidents or feedback, while the Trusted Contacts Integration ensures that users can effortlessly add contacts for emergency alerts. The Real-Time Navigation feature focuses on clear markers for safety hazards and alerts, and the Emergency Assistance & SOS feature offers quick, accessible help during high-stress situations. Additionally, the Plan and Book Safer Travel & Accommodation screen provides essential booking information with verified safety ratings, and the Buddy System feature connects solo travelers with verified local guides or fellow travelers for added security. Throughout, the wireframes utilize placeholder content and simple navigation paths to emphasize the functional aspects of each feature, ensuring a coherent and seamless experience for users.

As part of the design process for our safety-focused travel app for solo travelers, especially women, we created detailed wireframes using Figma. These wireframes visualize the core features and functionalities of the application, aiming to ensure a smooth, intuitive, and secure user journey from onboarding to post-trip feedback.

Use Cases Covered in Wireframes

The wireframes developed cover key use cases identified during user research and validated through insights and trends from both survey and interview phases. These use cases ensure a safer, more confident travel experience:

1. Dashboard
2. Incident Reporting / Post-Trip Feedback
3. Trusted Contacts Integration
4. Real-Time Navigation with Safety Alerts
5. Emergency Assistance & SOS
6. Plan and Book Safer Travel & Accommodation
7. Buddy System / Verified Local Guides

Each use case forms a part of the user journey and has been structurally mapped in the wireframes for clarity and alignment with our design goal.

SIGN IN/UP → DASHBOARD WIREFRAME

We started with a simple welcome screen that lets users immediately choose between logging in or signing up — nothing extra, just clear CTAs. Once they're in, the dashboard becomes the main hub. It shows their current location, user tier, and quick access to all core features like trip history or emergency info. The idea was to minimize taps and make it feel familiar — like any good travel or utility app — so users know where everything is right away.

Welcome Page

- A visually engaging entry point with a background image related to travel.
- Central logo with two main CTA buttons: Login and Signup.
- Simple, focused layout encourages onboarding.

Home / Landing Page

- Post-login screen where users land.
- Top shows Current Location, User status (Gold), and Profile icon.
- Below, a search bar with a hint.
- Five key sections are displayed as blocks — this makes navigation simple and intuitive.
- Bottom navigation bar (likely shared across all major screens).

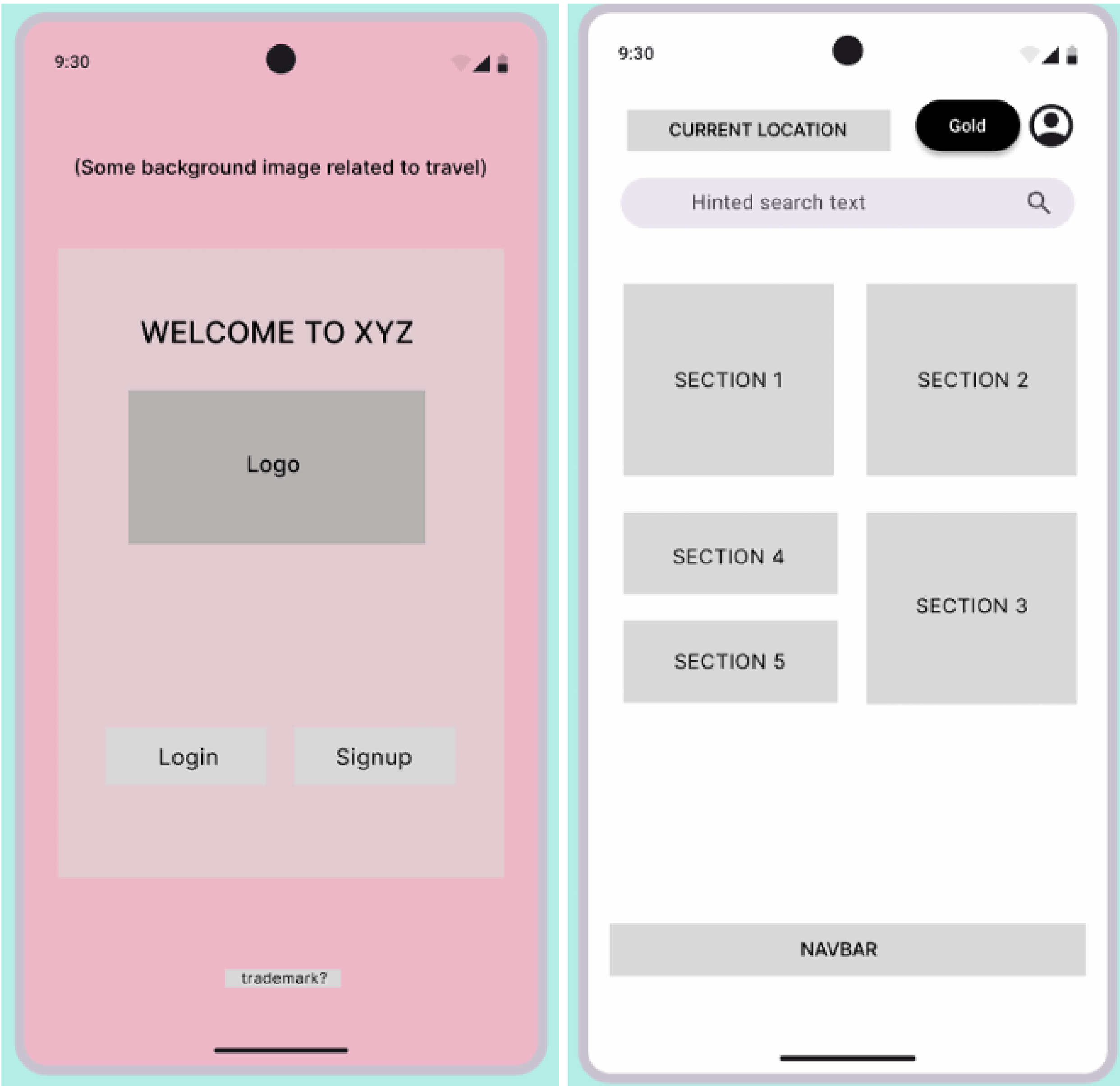


Figure 1: Signup and Home page

CASE-1: Customize the profile and Add Emergency Info

User Profile (Screen 1)

- Visual emphasis on user avatar.
- One-tap "Edit Profile" button.
- Profile settings are grouped functionally: emergency info, history, preferences, logout, etc.

Edit Profile Details (Screen 2)

- Fields to update name, email, phone, and address.
- Clean layout with a familiar mobile form feel.

This case ensures personalization and keeps essential information up-to-date.

Emergency Information (Screen 3)

- Displays the user's avatar for recognition.
- Sections for blood group, medical conditions, allergies, and address.

This part is critical for health safety, especially while traveling or during emergency.

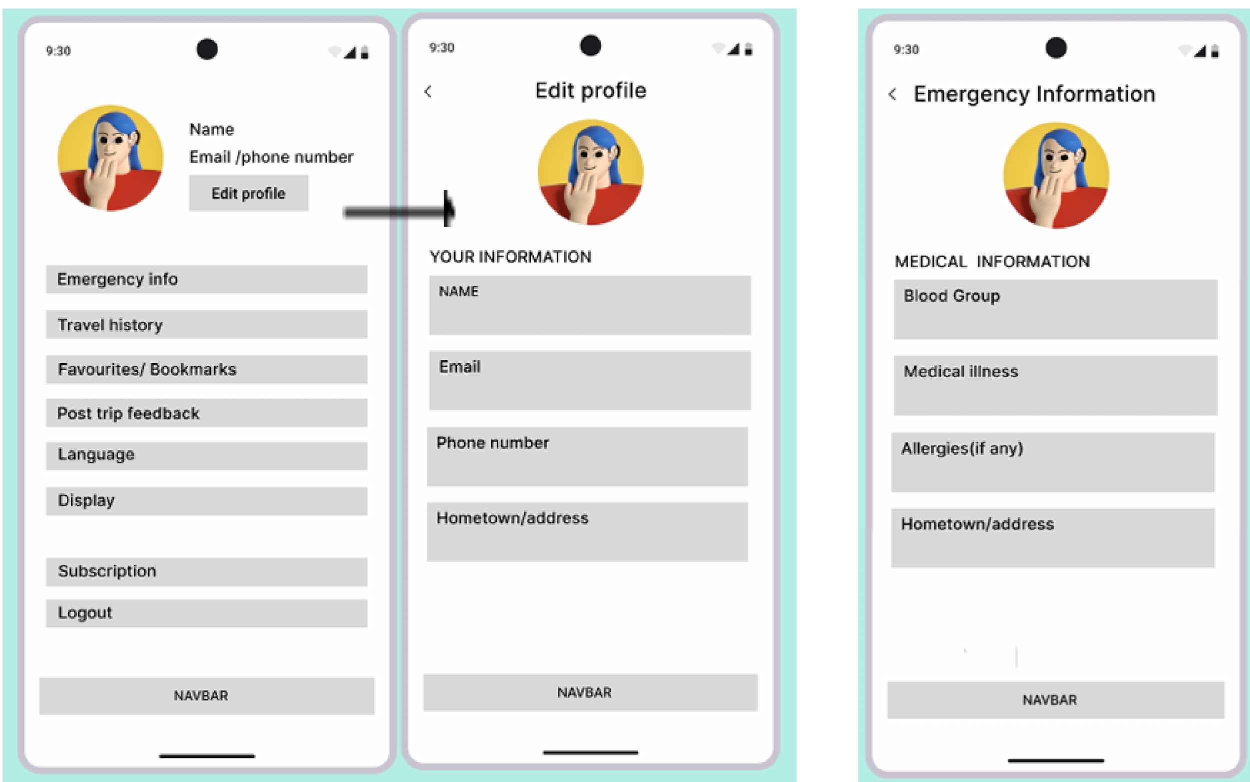


Figure 2: Customize the profile and Add Emergency Info

CASE 2 - Trusted Contacts Management

Trusted Contacts (Screen 2)

- Header with navigation and title.
- Buttons to Add new contact and Manage contact.
- List of trusted contacts with basic details.
- Each contact can be assigned permissions for emergency and notifications.

Add New Trusted Contact (Screen 3)

- Input fields for name and number.
- Toggleable permissions for emergency use and notification settings.
- Save/Cancel options ensure user control.

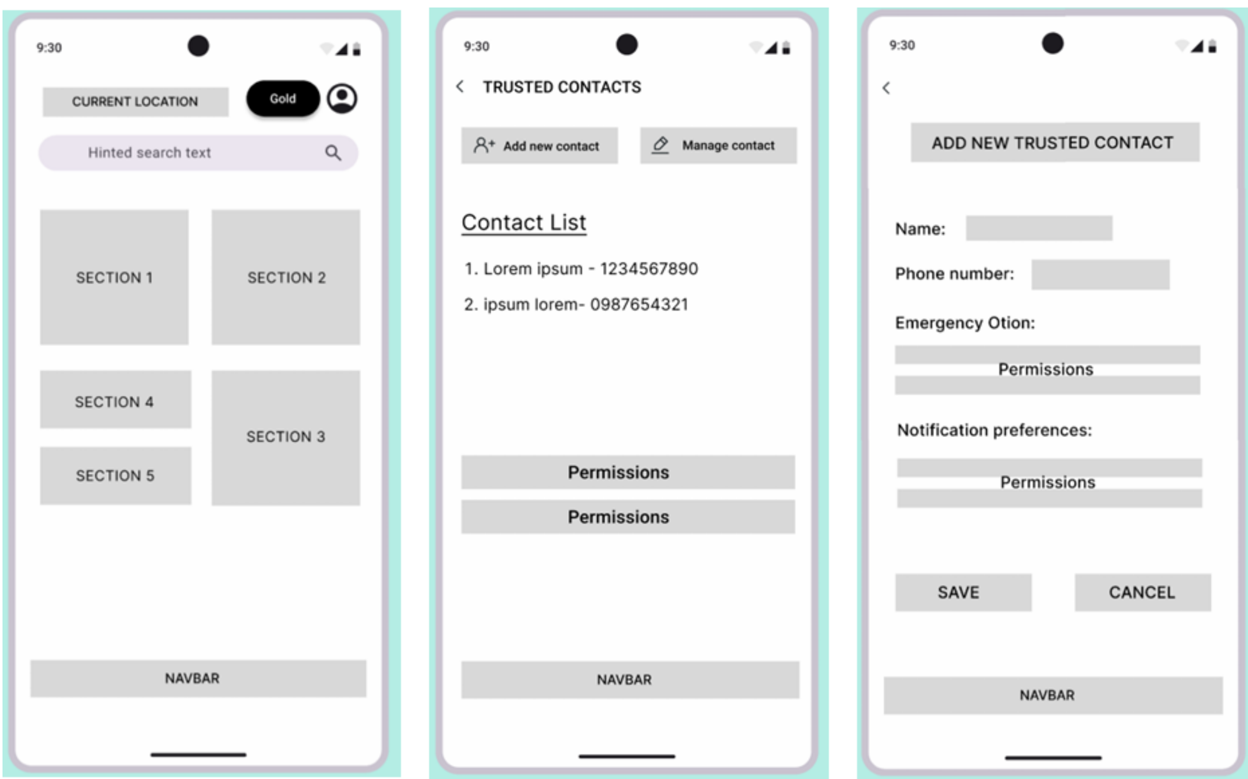


Figure 3: Trusted Contact Management

CASE 3 - Post Trip Feedbacks

Feedback List View

- Shows past trips (Trip 1, 2, 3) in a scrollable list format.
- Tabs like "Lorem" may represent filters or categories.

Trip Feedback Detail

- Dedicated screen per trip: includes image, star rating, review, and an incident report option.
- Action buttons for Discard and Submit at the bottom.
- This module encourages users to reflect on trips, rate them, and flag any incidents — helpful for platform quality and community safety.

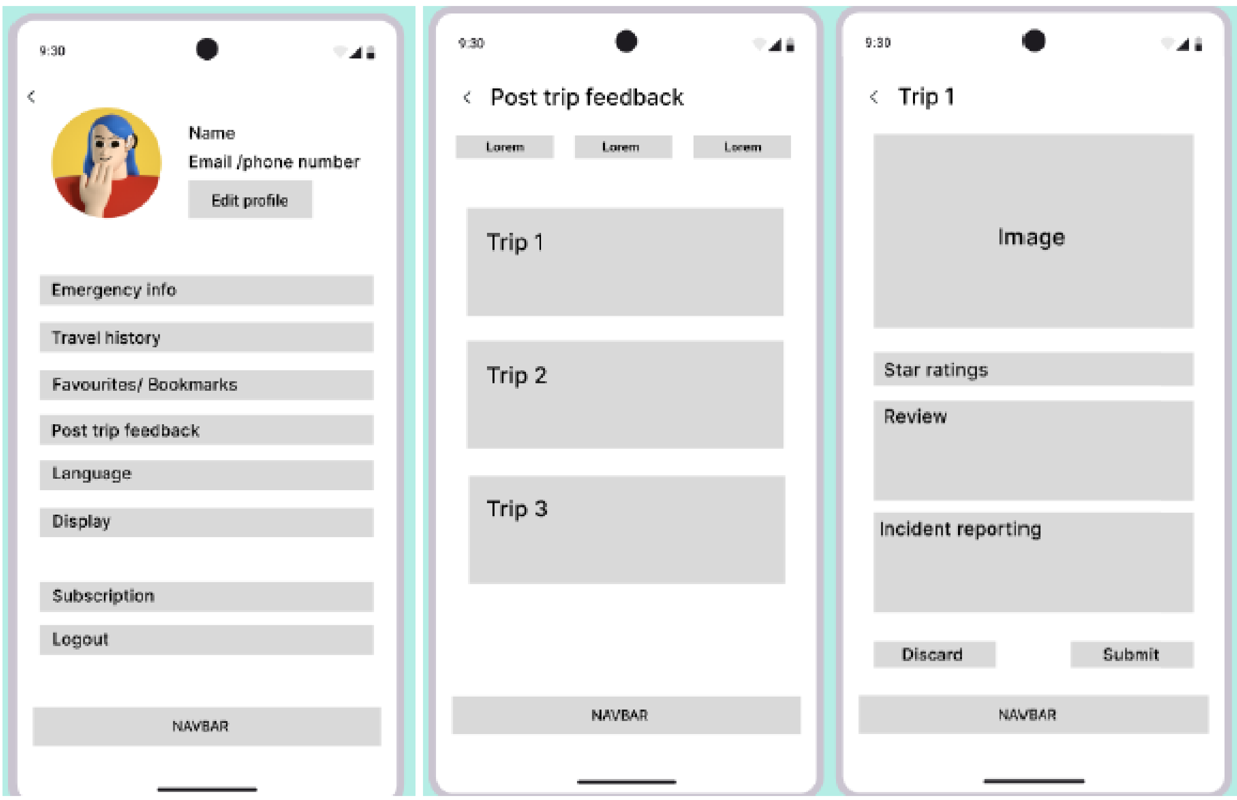


Figure 4: Post Trip Feedback

CASE 4: Discover and Assess Safety of Destinations

Search & Explore

- Search Bar to type in location or keyword.
- Filter button for advanced safety filters example Women Only, High Safety Rating etc.
- Trending Sections to highlight popular safe destinations.
- Category Thumbnails List of images that can be categorised by the user.
- Recommended Sections Similar to Trending Section to highlight popular safe destinations.

Category & Destination Title

- List of Places each showing an image, brief description, and safety rating or badge.
- User Flow: Tapping any place leads to the destination detail screen.

Destination Detail

- Carousel (Main Image) placeholder at the top.
- Star Ratings to view risk level.
- Tabs for Description, Review, Photo, and Location.
- Description: Basic info, key attractions, any safety advisories.
- Review: User-generated safety reviews or official verification.
- Photo: Gallery type for reference.
- Location: Map or address details.

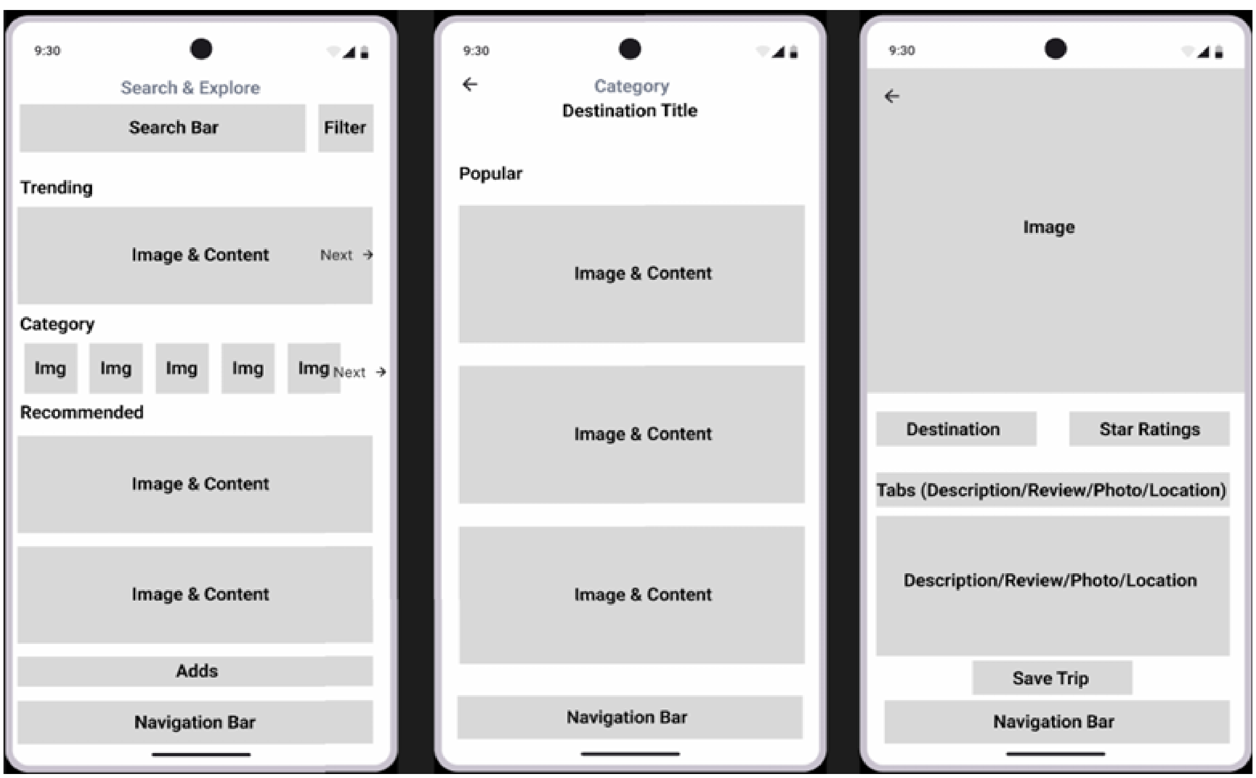


Figure 5: Search and Explore

CASE 5: Buddy System Matching

Find a Buddy

- Filter Options: Verified Only, Nearest Distance, Interests, Women Only, Willing to Accept.
- Buddy List:
  - Profile Photo Placeholder & Badges: Shows whether the user is verified, plus any relevant tags like Women Only.
  - User Description: Brief text about the buddy's interests or travel style.
  - Send Request Button: Allows the user to initiate a buddy request.

Buddy Connections

- Request / Sent: Shows buddies you've sent requests to or received requests from.
- Tick / Cross buttons to accept or decline a request.
- Your Connections: A list of currently connected buddies with profile photos, badges, and short descriptions.

Chat System

- Header: Displays buddy's name or status.
- Conversation Area: My Message / His Message for communicating.
- Message Input Field:
  - A text box to write the message.
  - Send Button to submit the chat message.

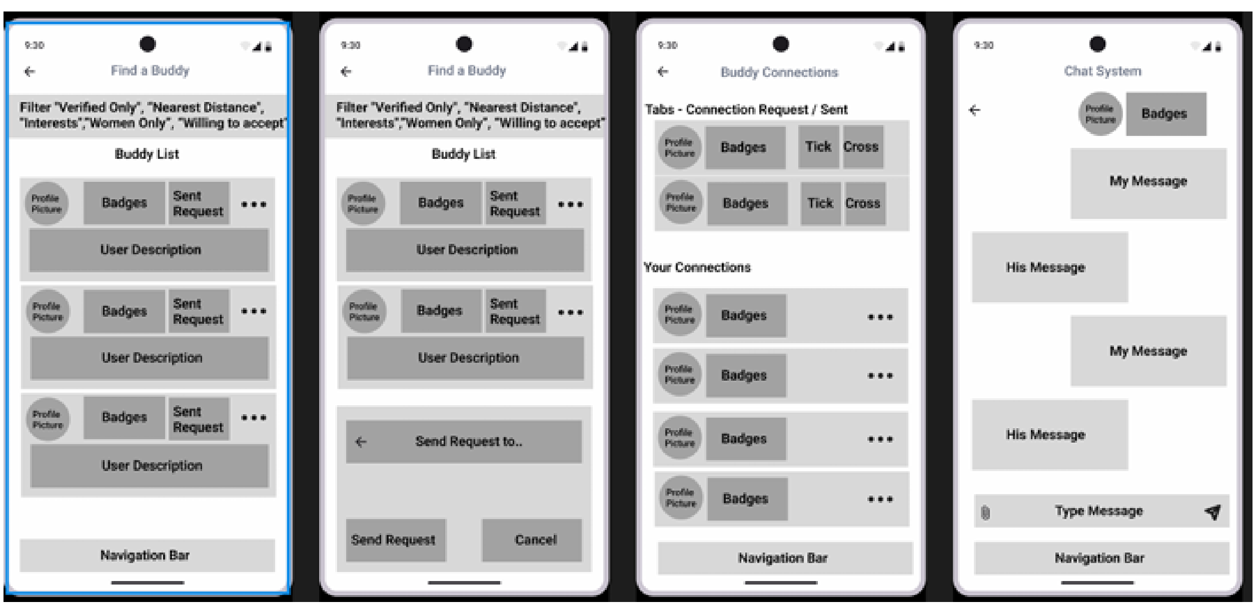


Figure 6: Buddy System Matching

CASE 6: Emergency Assistance and SOS

Emergency SOS

- Cancel button at the top to abort
- Countdown Timer:
  - Big Numbers being displayed before alerting emergency services like countdown.
  - Instructional Text: Hold to activate emergency alert for user acknowledgement.
- SOS Button: A prominent red circle called SOS for emergency.

Emergency Call Initiated

- Cancel button remains available to end the process.
- Animated Call Symbol for the app is connecting to emergency services. It gives user some relief while looking to call spinner.
- Buttons:
  - Add Incident Details: Option to quickly note what's happening.
  - Share Live Location: Automatic and optional, sharing location everytime the button is clicked to emergency contacts.

SOS Preferences

- Toggles
  - Auto-Call 911 or local helpline.
  - Share GPS with Contacts.
- List of Contacts: Telling user which contacts to include during SOS emergency.

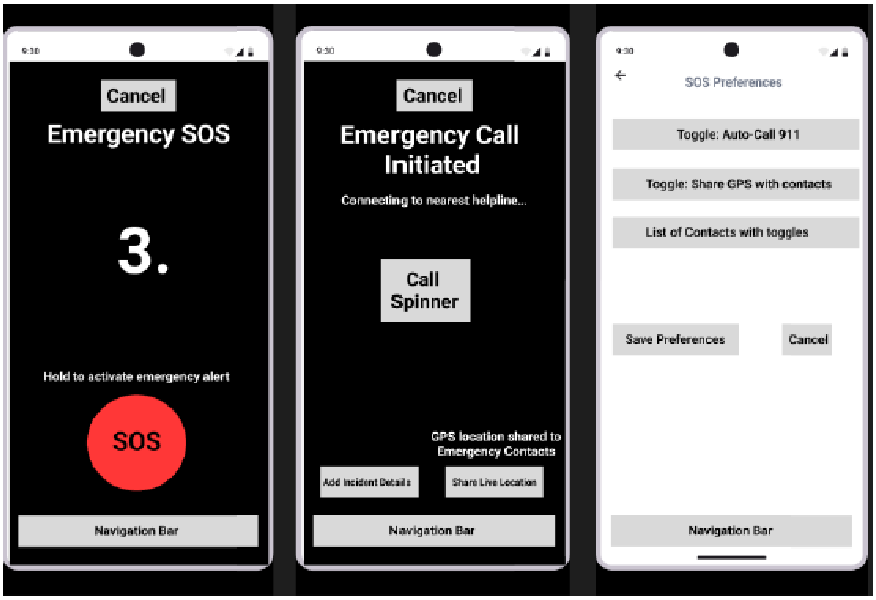


Figure 7: Emergency Assistance and SOS

CASE 7: Real-Time Navigation with Safety Alerts

Search Location

- Search Bar / Field: For entering a location or choosing a destination.
- Vehicles While Traveling: Dropdown of vehicle currently in use for travel.
- Real-Time Alerts: A placeholder for safety messages that appear based on location.
- SOS Button: A prominent red circle for emergencies.
- Map View: Map showing user location and routes.

Your Location / Choose Destination / Categorize Vehicles

- Your Location Panel: A button indicating current location.
- Choose Destination: A text field to select from a list of addresses.
- Categorize Vehicles: A placeholder section where users can pick vehicle type.
- Address History: A list of previously used addresses for quick selection.

Map with Real-Time Alerts

- Map View: Map showing user location and routes.
- Current Location Magnifier / Nearby Location: Buttons that zoom to user location or show nearby safe spots.
- Real-Time Alerts Bar that displays caution messages.
- SOS Button: Large red circle for instant emergency access.

Share Live Location

- Duration Toggle: Options like 15 minutes, 1 hour etc.
- Add Message: Note optional for brief text.
- List of Nearby Places: User can share nearby meetup places.

Report Incident

- Enter Location for Incident: A field or button to specify exact spot.
- Classifying Incidents: Categories for classifying all incidents.
- Describe the Incident: A text box for details.
- Submit / Cancel Buttons: Confirm or discard the report.

List of Routes, Distance and User Safety

- Map: Small map to continue to show user that he is still in navigation screen.
- Route Options Panel (Bottom): A list of routes with time, distance, and a safety rating.
- Select Route: A card tap that sets the chosen path.

Lorem Ipsum (List of People)

- Map: Small map to continue to show user that he is still in navigation screen.
- Profile Picture / Name of Person: To display particular person to share location.

Figure 8: Real-Time Navigation with Safety Alerts 1

Figure 9: Real-Time Navigation with Safety Alerts 2

9.2.1 Storyboard

The overall theme of our app revolves around a **blue and orange color palette**, carefully chosen for both aesthetic appeal and functional effectiveness. **Blue** represents trust, safety, and calmness—qualities that are crucial for solo travelers, particularly women, as they navigate unfamiliar places and rely on safety features. It promotes a sense of security and reliability, which is why we've used it prominently across the app.
Orange, by contrast, is an energizing color that incites action. It's used deliberately for buttons and alerting, directing people's attention to salient tasks such as emergency actions, alerts, and navigating. The blue and orange contrast prevents users from missing calls-to-action without becoming overwhelmed.
As a whole, the color scheme of blue and orange brings about a well-balanced contrast between trust and urgency, a friendly and emotional experience that induces engagement, usability, and composure in what could be a stressful situation. The **SOS screen** follows a bold and urgent design, utilizing a **red button** on a **black background**, which is a deliberate choice to evoke a sense of urgency and alertness. Red, as a color, is universally associated with emergency situations, danger, and the need for immediate action, making it the ideal choice for the SOS button. Its high contrast against the black background ensures that the button is easily visible, even in high-stress scenarios where quick action is critical.
The choice of a black background serves to isolate the SOS button from any distractions, making it the focal point of the screen. The red button stands out clearly, instantly guiding the user's attention to the emergency feature. This design ensures that in a moment of panic or urgency, the user will instinctively know where to tap to initiate a response.
In contrast to the rest of the app's white background with blue and orange elements, this bold design on the SOS screen ensures that users can quickly and easily access the emergency feature without confusion. The visual simplicity and clarity of the screen help to minimize stress and provide a direct route to getting help when needed.

We have designed a travel safety application designed especially for solo travellers. The storyboard visually captures the user journey through key interactions within the app, allowing us to examine both the design flow and the overall user experience.

Use Cases Covered in Storyboard

1. Customise the profile and add emergency info
2. Trusted contacts management
3. Post Trip Feedbacks
4. Discover and Assess Safety of Destinations
5. Emergency Assistance & SOS
6. Buddy System / Verified Local Guides
7. Real-Time Navigation with Safety Alerts

By following this storyboard, our stakeholders can gain a clear understanding of how our design supports the user's journey, improves usability, and enhances overall safety during travel.

User Case-1: Customize the profile and add emergency info

This use case allows users to update and personalize their profile information including basic details like name, avatar, bio and essential emergency information like emergency contacts,medical details, special instructions.It also allows users to control what data is shared in emergency scenarios and ensures that trusted contacts have accurate information.

Key Actions:

- Edit personal details like full name, email, and phone.
- Update emergency contacts and input medical information.
- Save changes and receive confirmation that the profile and emergency info are updated.

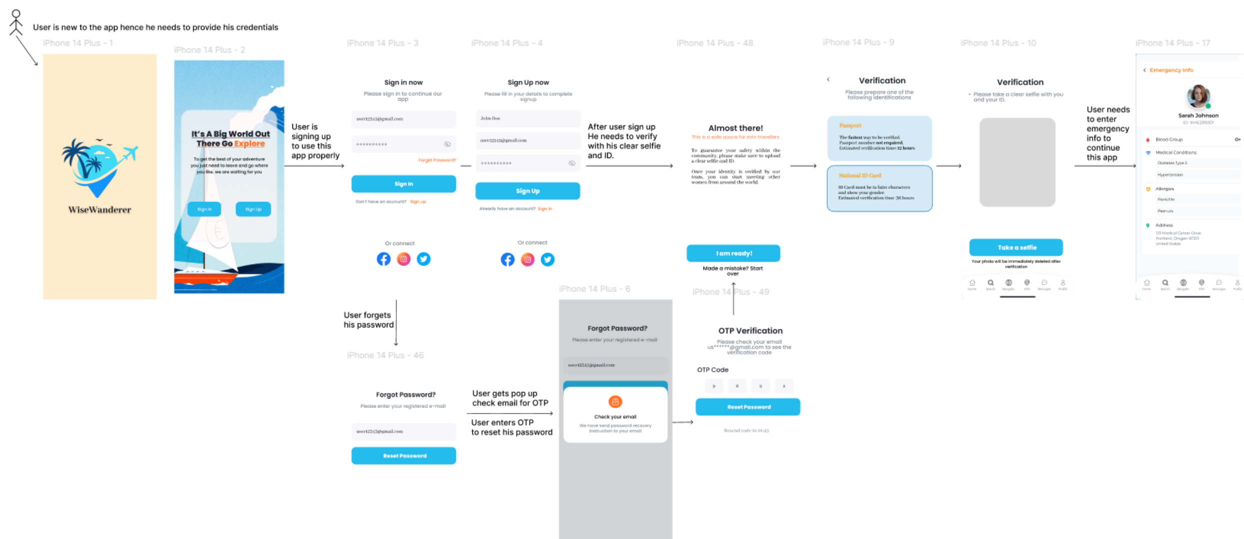


Figure 10: Customize the profile and add emergency info

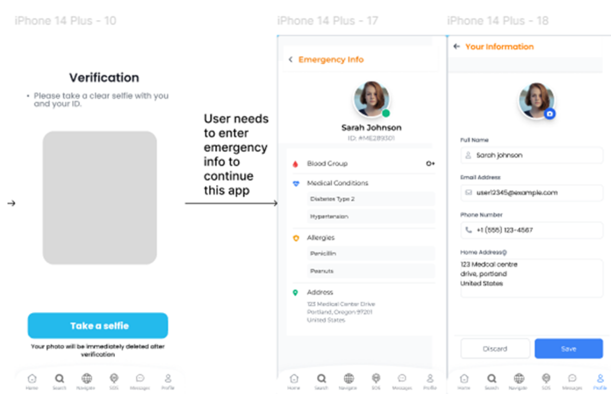


Figure 11: Customize the profile and add emergency info

Use Case 2: Trusted Contacts Management

This use case focuses on managing a list of trusted contacts who can receive your location updates and SOS alerts. It ensures that users can quickly add, edit, or remove contacts, and control permissions regarding who gets notified in emergencies.

Key Actions:

- View a list of pre-approved contacts along with their verification status.
- Add new contacts using a form with fields for name, phone number, and relationship.
- Toggle options for sharing live location, emergency SOS, or geofence alerts.
- Update or remove existing trusted contacts with clear confirmation prompts.

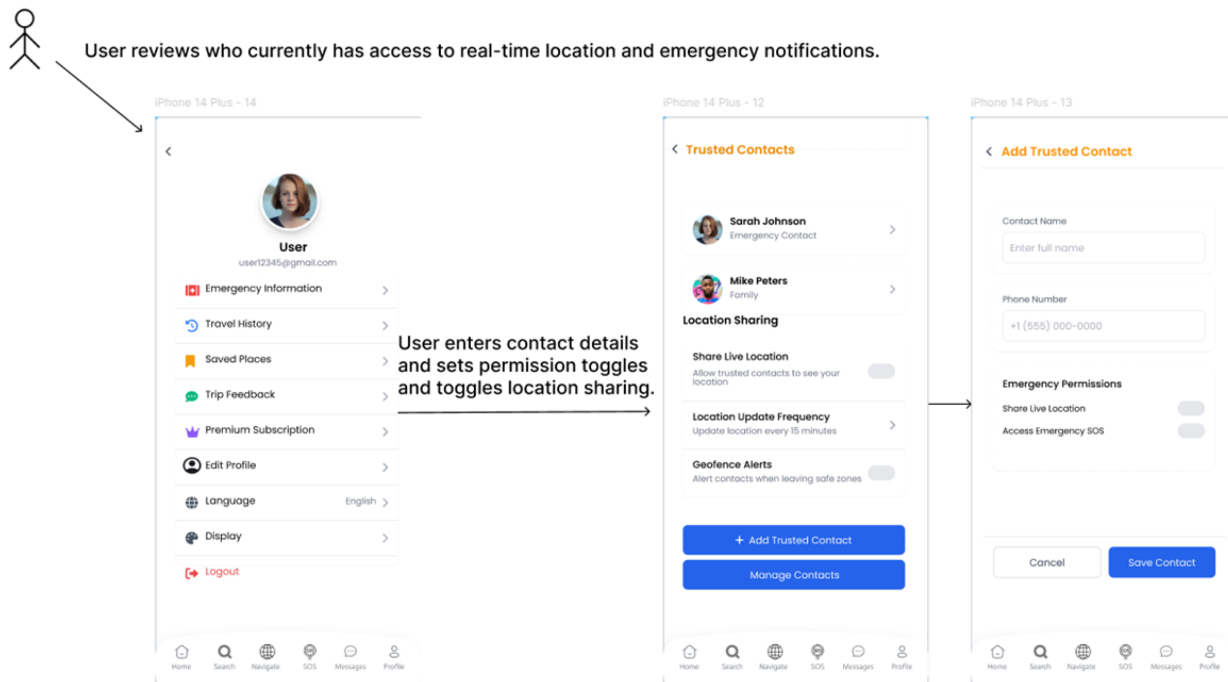


Figure 12: Trusted Contacts Management

Use Case 3: Post Trip Feedback

This use case is all about capturing feedback from users after they complete a trip. Users can rate the safety and overall experience of a destination,leave comments,and even upload supporting images.

Key Actions:

- Display a list of completed trips with an option to tap and leave feedback.
- Provide a detailed feedback form where users select a star rating, enter text comments, and attach photos or additional details if necessary.
- Submit feedback to update the trip status and inform the community through reviews.

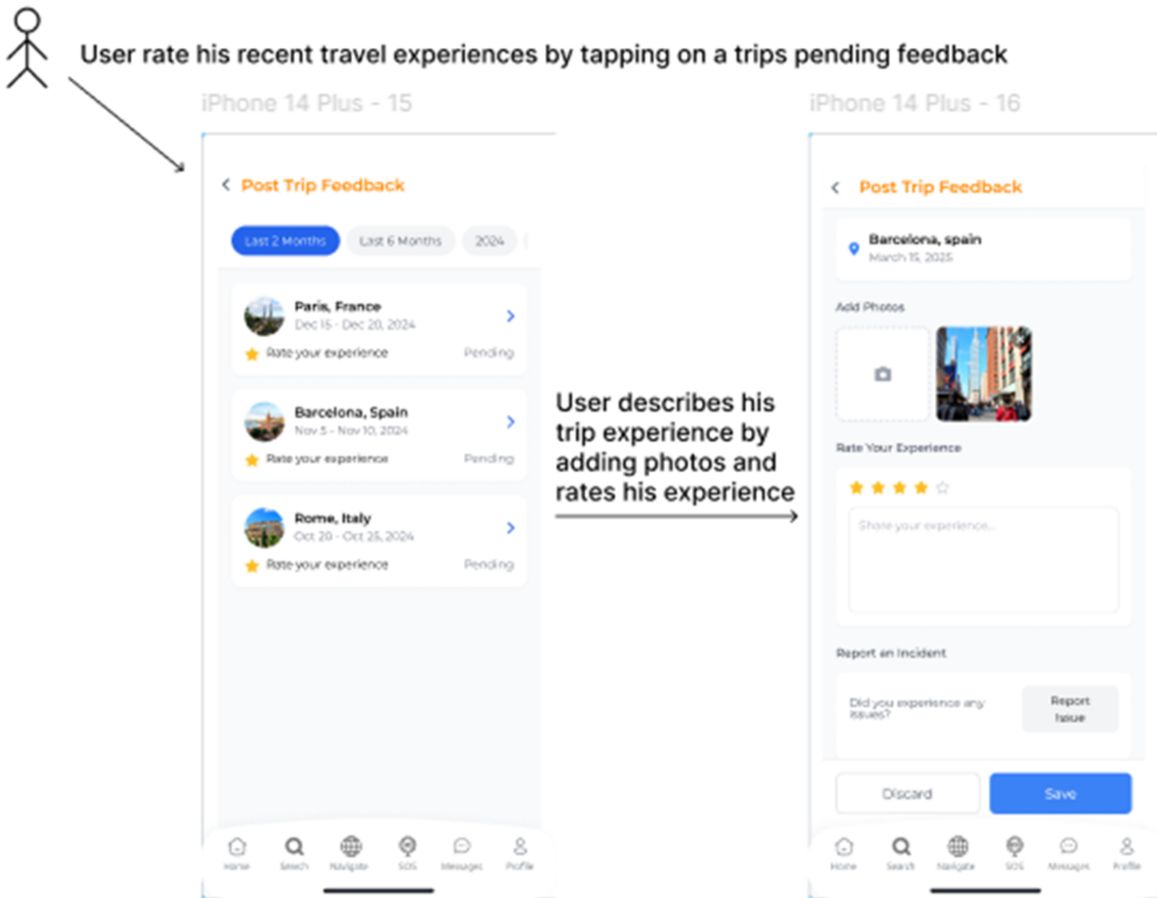


Figure 13: Post Trip Feedback

Use Case 4: Discover & Assess Safety of Destinations

This use case enables users to explore different travel destinations and assess their safety through safety scores, user reviews, and visual content. It helps users make informed travel decisions based on safety data.

Key Actions:

- Browse trending or recommended destinations via a search and category-based interface.
- Tap on destination cards to view detailed information, including images, safety scores, verified reviews, and maps.

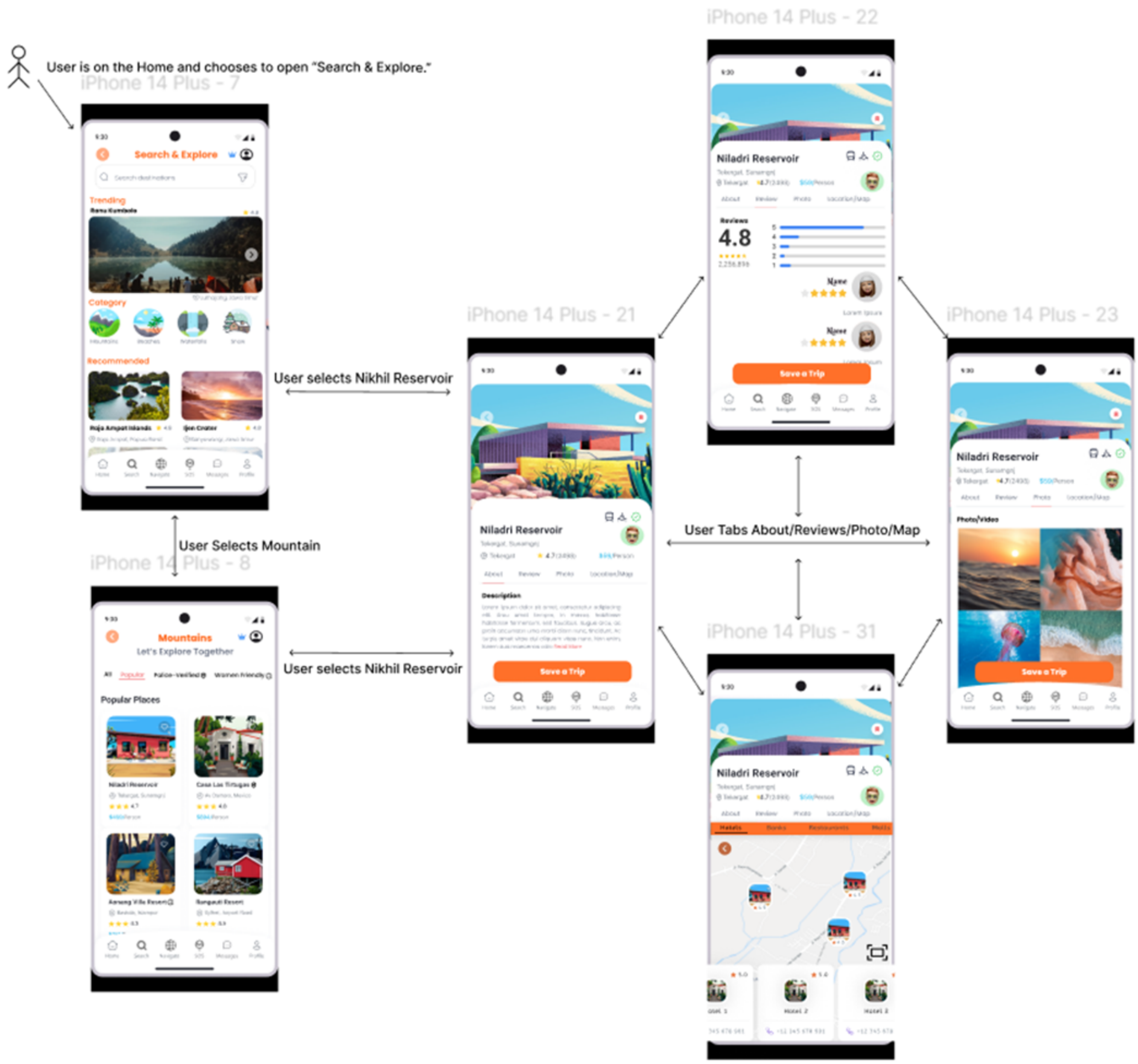


Figure 14: Discover & Assess Safety of Destinations

Use Case 5: Buddy System

This use case allows solo travelers to connect with verified travel buddies for companionship and shared safety on their trips.It facilitates discovering potential buddies, sending connection requests, and initiating chats.

Key Actions:

- Browse a list of travel buddy profiles with basic info, avatars, and verification badges.
- Send a buddy request by tapping on a profile.
- Manage buddy requests like accept, reject, or view pending connections.
- Start a chat with connected buddies to coordinate travel plans and provide mutual support.

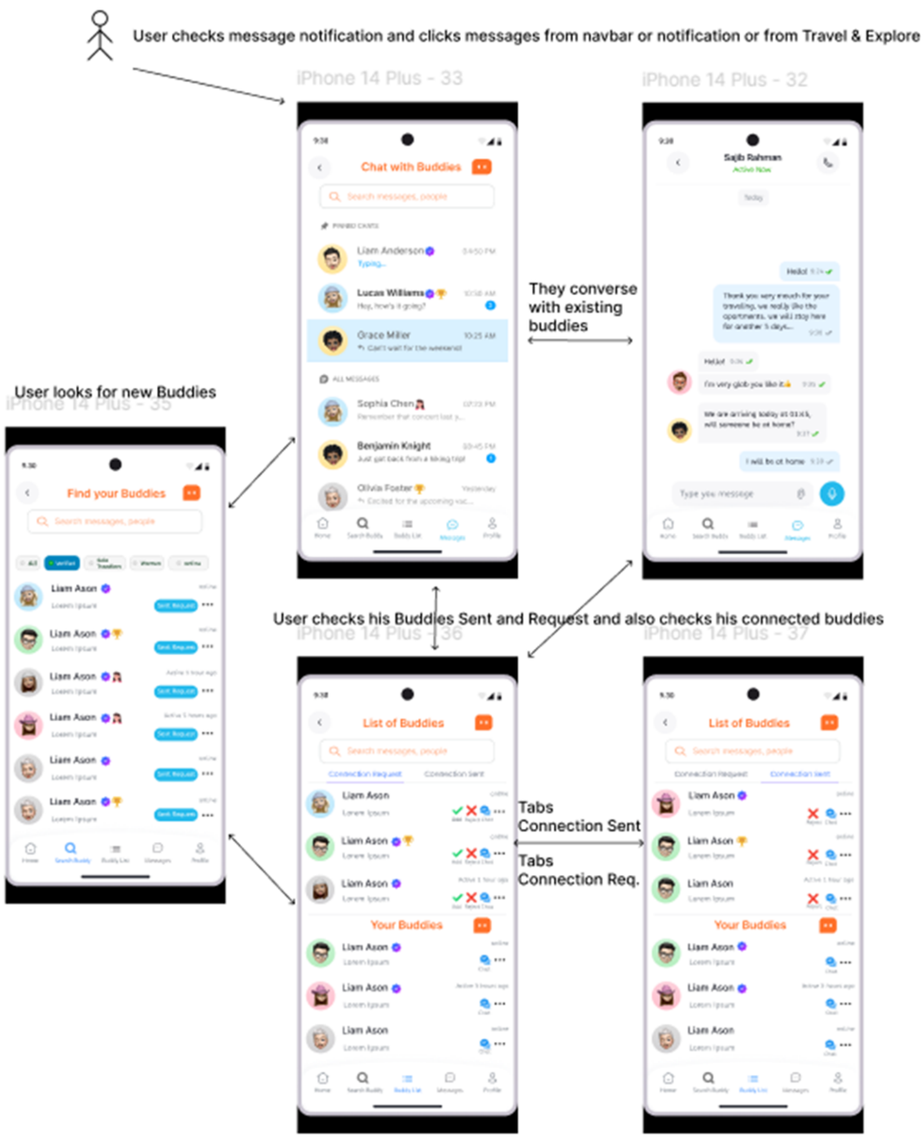


Figure 15: Buddy System

Use Case 6: Emergency Assistance & SOS

It provides immediate emergency support for users in distress. When a user feels unsafe or faces an urgent situation, the Emergency Assistance & SOS feature enables them to quickly initiate an alert.

Key Actions:

- Display a prominent, large red SOS button and shows countdown timer while the user holds the button.
- Once the countdown completes, automatically trigger the emergency alert and initiate a call to local emergency services.
- Immediately share the user's live location with designated emergency contacts.

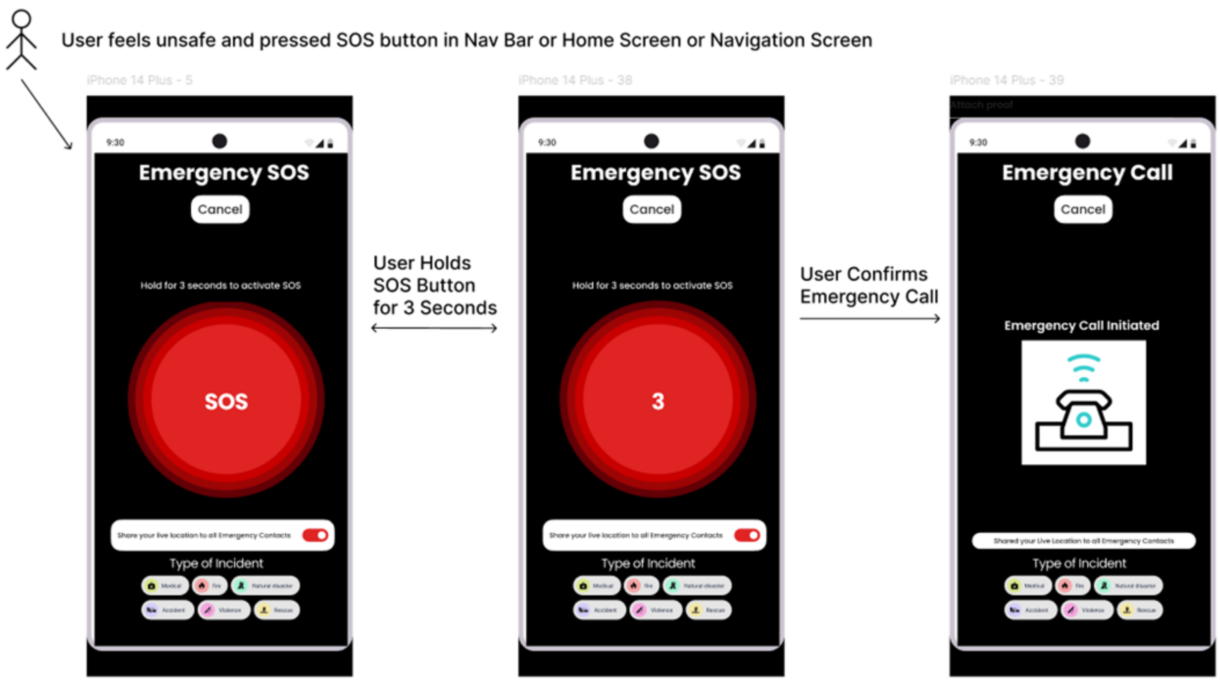


Figure 16: Emergency Assistance & SOS

Use Case 7: Real-Time Navigation with Safety Alerts

This use case provides live navigation functionality that not only directs users via a map but also integrates real-time safety alerts. It helps users safely reach their destinations by showing route data and warning about potential incidents along the way.

Key Actions:

- Display a live map with the current location, route, and distance/time remaining.
- Show prominent safety alerts e.g., caution banners for areas with recent incidents.
- Allow additional actions like sharing live location, reporting an incident, or engaging with emergency features (like an SOS button).

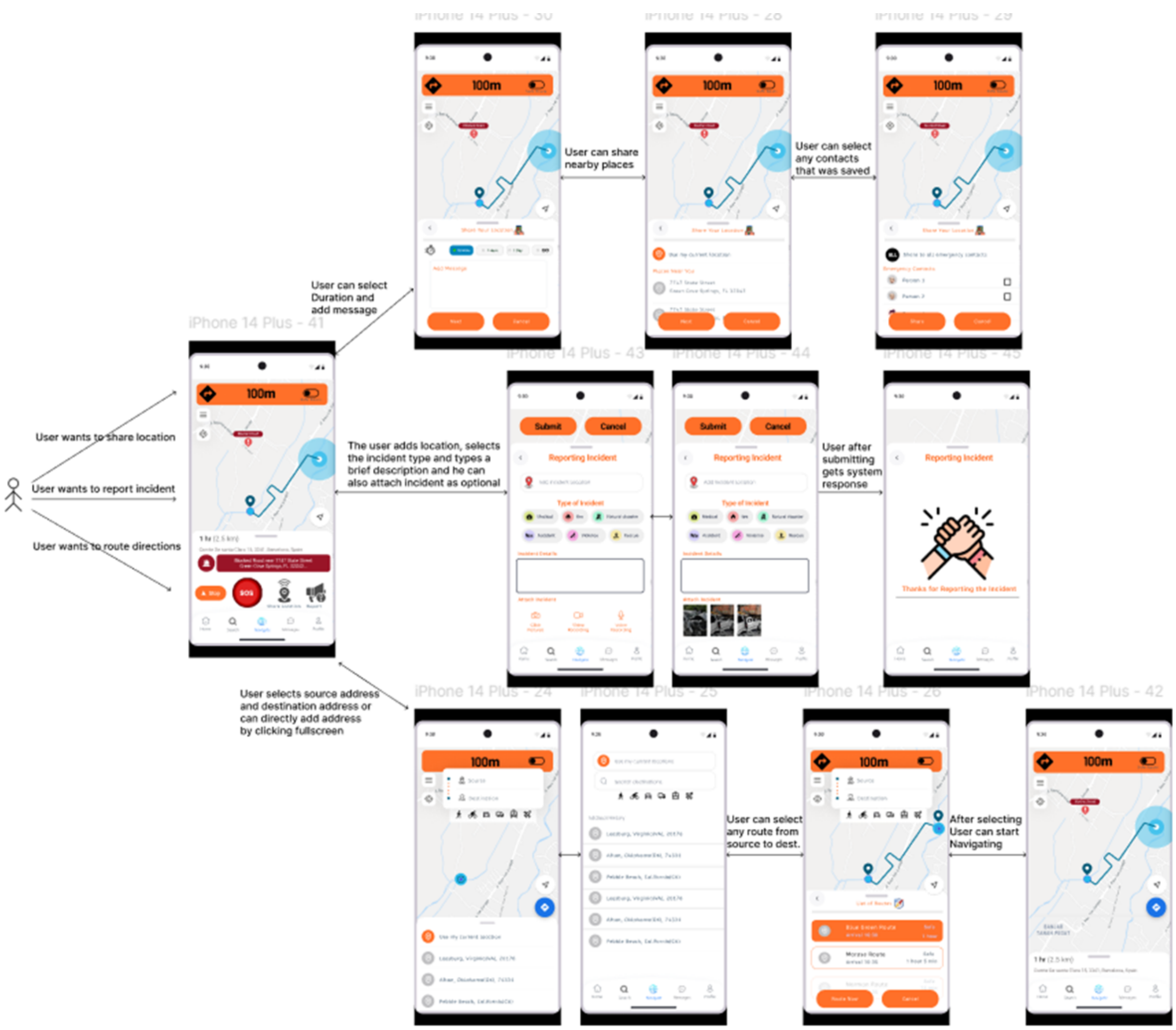


Figure 17: Real-Time Navigation with Safety Alerts

9.3 Prototype

Our prototype shows the end to end functionality and user flow of the our application especially for solo travelers.Created using Figma, the interactive prototype brings together the finalized wireframes and storyboards into a simulated and functional user experience. Each interaction in the prototype corresponds directly to use cases used during our design process. These include login/sign-up flows, profile customization, safety checks for destinations, SOS systems, buddy networks, and incident reporting.

Link to Prototype: Click here to view the prototype

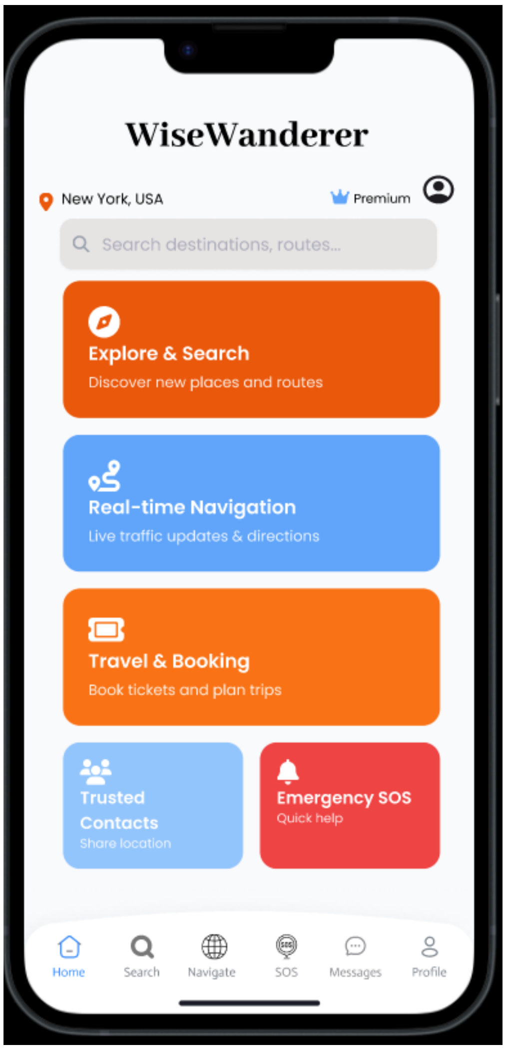


Figure 18: Home Page

Home Page of App

Explore & Search – Use Case 4 - Discover & Assess Safety of Destinations which navigates into destination's detail page.
Real Time Navigation – Use Case 7 - opening the map view where you select source/destination, choose routes, share location and report incidents.
Travel & Booking – Use Case 5 – Provides user buddy system where user can chat with friends, add connections and search other users.
Trusted Contacts – Use Case 2 - Showing list of contacts, can add, edit, and set permissions for who sees your live location.
Emergency SOS – Use Case 6 - the large red prominent SOS button screen and countdown screen for holding and triggering an immediate help call.

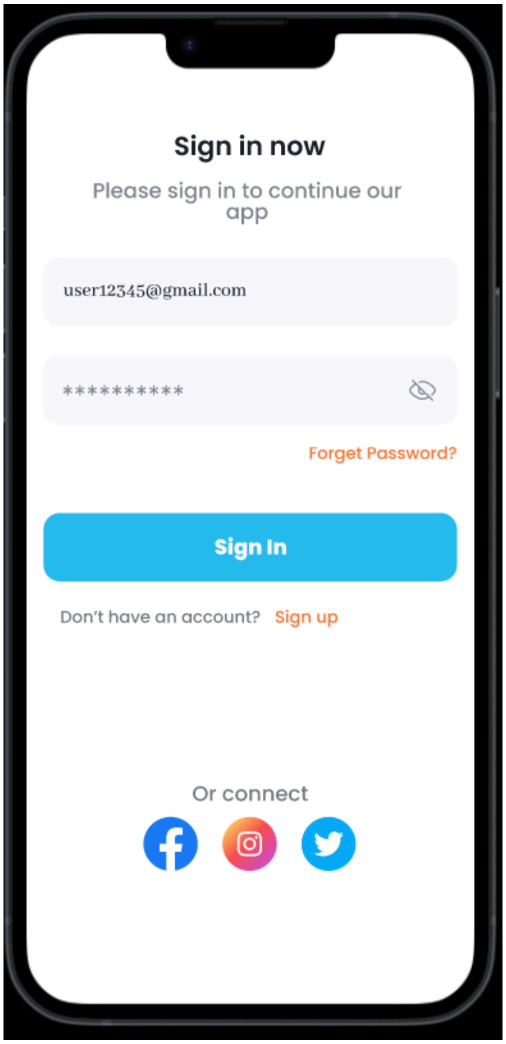


Figure 19: Profile and Emergency Info

Use Case 1: Customise the Profile and Add Emergency Info

Sign in – Allows user to login and move to Home Page.
Forgot Password – Asks for email, confirms OTP, then navigates to Home Page.
Sign Up – Requires verification via Passport/National ID and a selfie before granting access.
After sign up, user will be asked to verify himself. Click Passport or National ID Card which navigates to a screen that displays scan image. By clicking "Take a Selfie" user will move to Home Page.
After sign up user will be asked to verify himself. Click Passport or National ID Card which navigates to a screen that displays scan image. By clicking "Take a Selfie" user will move to Home Page.

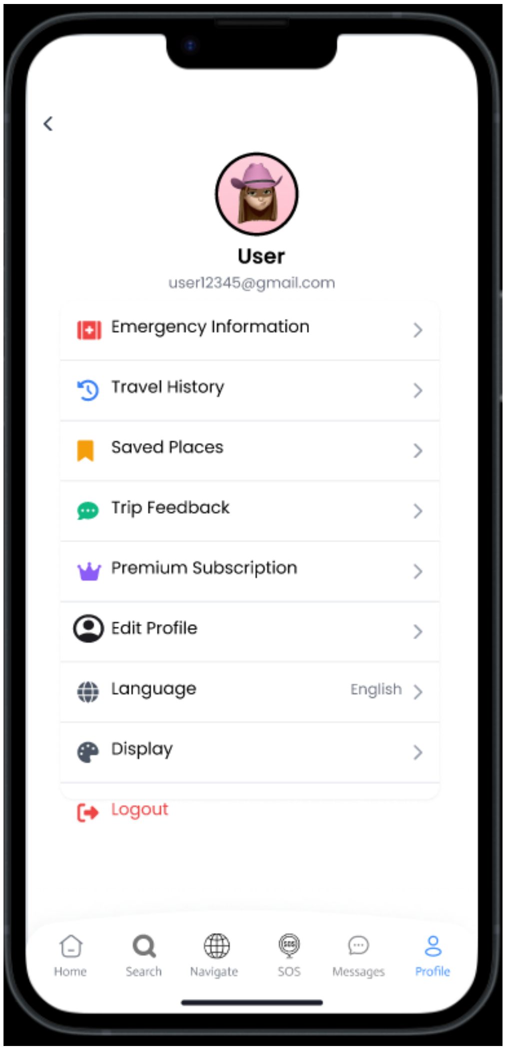


Figure 20: Profile Options and Emergency Info

Use Case 1 (continued): Profile Options and Emergency Info

By tapping the profile icon in the navbar, user can open their profile.
User can select:

- **Emergency Information**
- **Travel History**
- **Trip Feedback**
- **Edit Profile**
- **Logout**

Selecting **Emergency Information** opens a dedicated form to enter emergency contacts, medical conditions, and instructions. After filling all the details, the app confirms the input.
Selecting the **Edit** button brings up a Personal Details form where user can enter name, email address, phone number, and home address. The user can then save these details.

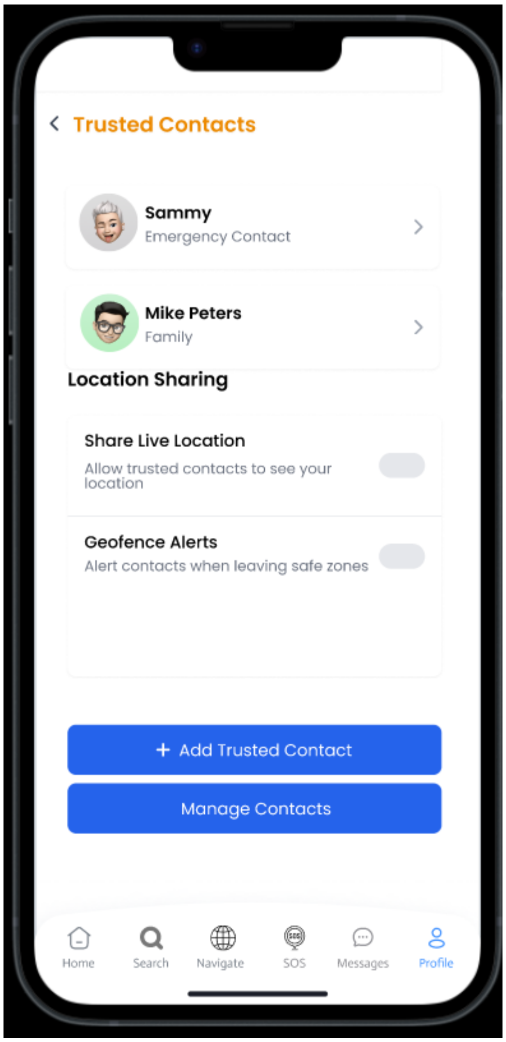


Figure 21: Trusted Contacts Screen

Use Case 2: Trusted Contacts Management

This screen allows users to manage their list of trusted contacts who can view their live location.
User can:

- View the list of existing trusted contacts
- Add a new contact by entering name and phone number
- Edit contact details
- Delete a contact from the list

Each contact can be toggled to enable or disable live location sharing permissions. The interface ensures that only verified trusted individuals can access user location.

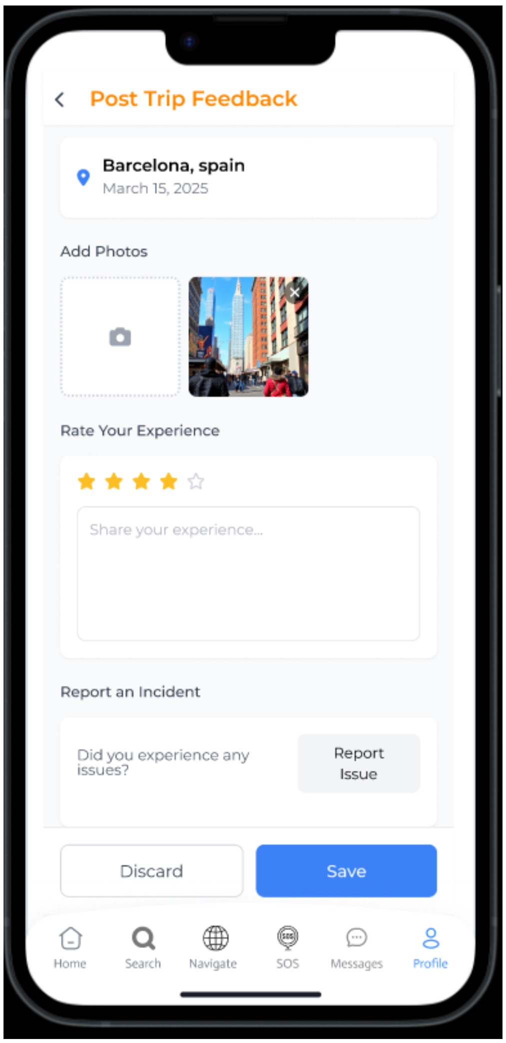


Figure 22: Post Trip Feedback Screen

Use Case 3: Post Trip Feedback

After completing a trip, users are prompted to share feedback based on their travel experience.
The screen includes:

- A rating scale from 1 to 5 for trip satisfaction
- Open text field to write suggestions or highlight issues
- Submit button to confirm feedback

Feedback collected is used to enhance future experiences and ensure continuous improvement in safety, usability, and satisfaction.

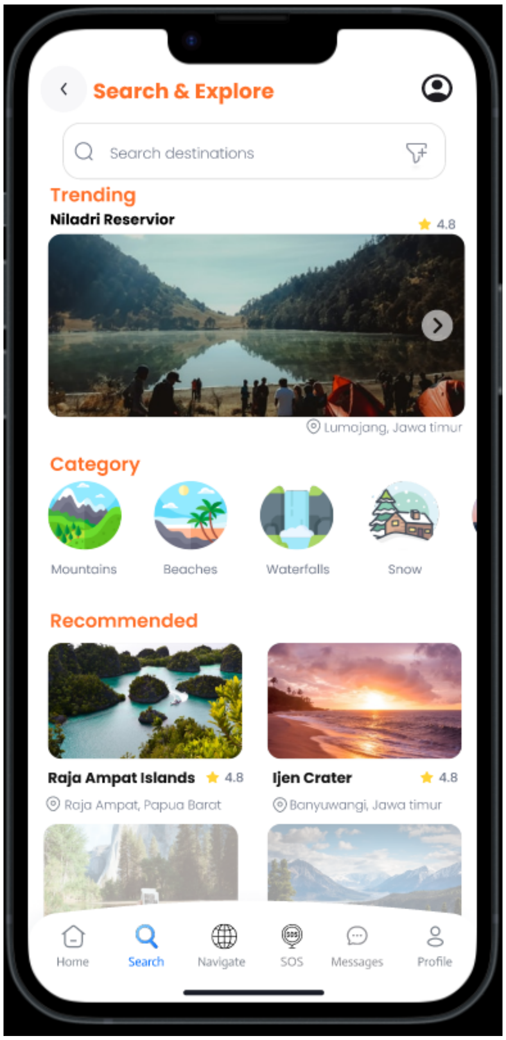


Figure 23: Discover Destinations

Use Case 4: Discover & Assess Safety of Destinations

This screen enables users to search and explore travel destinations, along with their safety indicators.
User can:

- Enter destination keywords in the search bar
- View a list of matching locations
- Tap on a destination to see its detailed safety profile

Each destination profile provides information on local safety levels, common incidents, crowd density, and reviews from other travelers. This helps users make informed decisions before selecting a place to visit.

Figure 24: Place Detail Screen

Use Case 4 (continued): Place Detail View

Once a destination is selected, the app displays detailed safety information about that place.
The Place Detail screen contains:

- Description of the location and key attractions
- User reviews and safety ratings
- Photo gallery of the area
- Map view highlighting nearby hotels, malls, and hospitals

This screen empowers travelers to evaluate risk factors and plan their visit based on verified insights and community feedback.

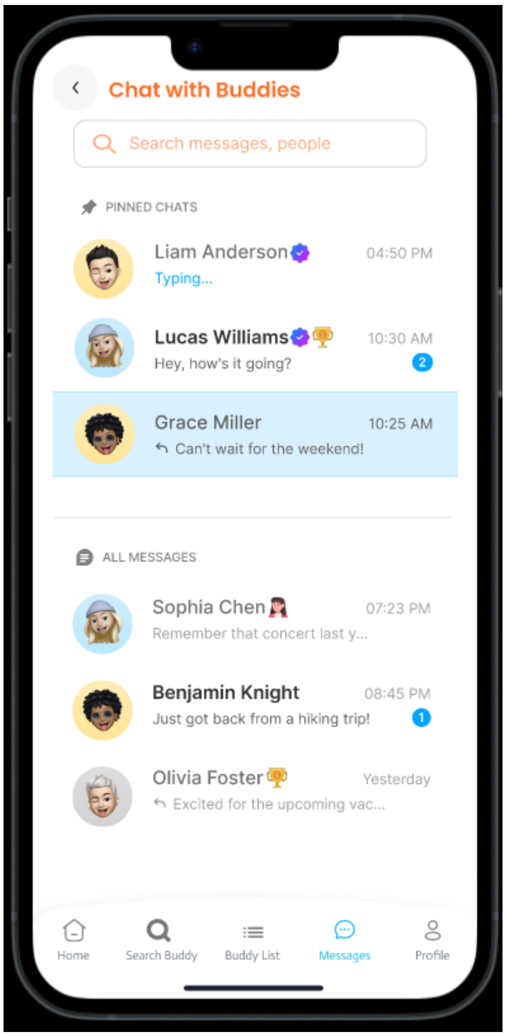


Figure 25: Buddy System Interface

Use Case 5: Buddy System (Search, Connect, Chat)

This feature allows solo travelers to connect with other trusted users for companionship and added safety.

Search Buddy:

- Displays a scrollable list of verified users with brief bios
- Users can apply filters like gender, distance, and interest tags
- Tapping "Send Request" sends a buddy request to the selected user

Connection Management & Chat:

- Received and sent requests are shown under "Connection Requests" and "Requests Sent"
- Users can accept or reject requests using tick and cross icons
- Once connected, users can access a basic chat interface to communicate and coordinate

The buddy system ensures a safer journey by allowing real-time human connection based on preferences and verified profiles.

Figure 26: Emergency Assistance and SOS

Use Case 6: Emergency Assistance & SOS

This feature allows users to immediately raise an alert in emergency situations.

SOS Trigger:

- A large red SOS button appears prominently across the app
- User must press and hold the button for 3 seconds to confirm it's not accidental
- A cancel option is also available to abort the process

Emergency Call Initiation:

- The app places a call to local emergency services
- The user's live location is shared with pre-set trusted contacts
- Additional options like adding incident details are presented

This mechanism ensures fast, reliable, and non-intrusive emergency support for solo travelers in distress.

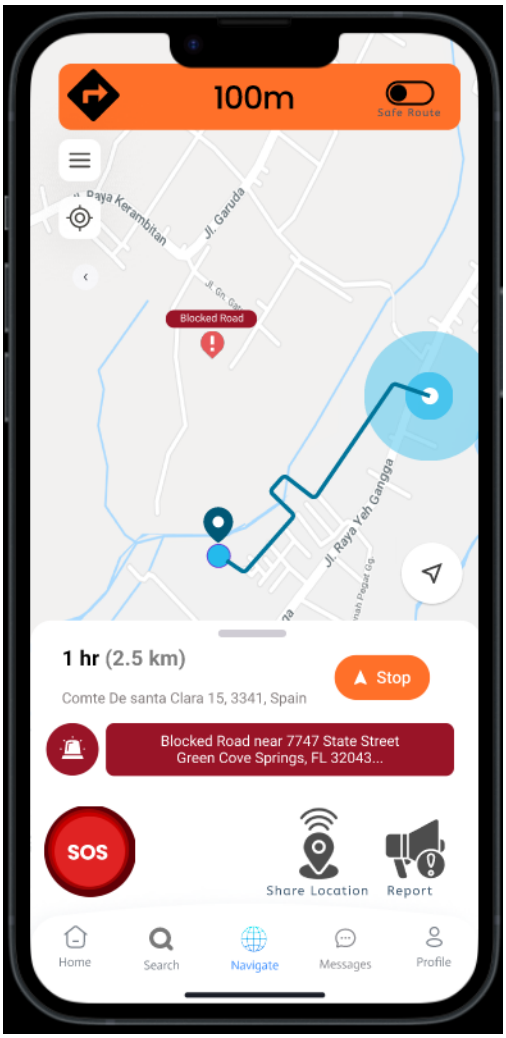


Figure 27: Real-Time Navigation – Map Setup

Use Case 7: Real-Time Navigation with Safety Alerts

Part 1 – Map Setup and Choosing Location

The navigation system allows users to plan a safe route using real-time data.

- Tap on "Real-Time Navigation" to access map view
- Tap the parallel-line icon to enter source and destination
- Users can choose locations from a suggestion list or enter manually

The setup ensures flexibility and quick navigation from current location to a desired destination.

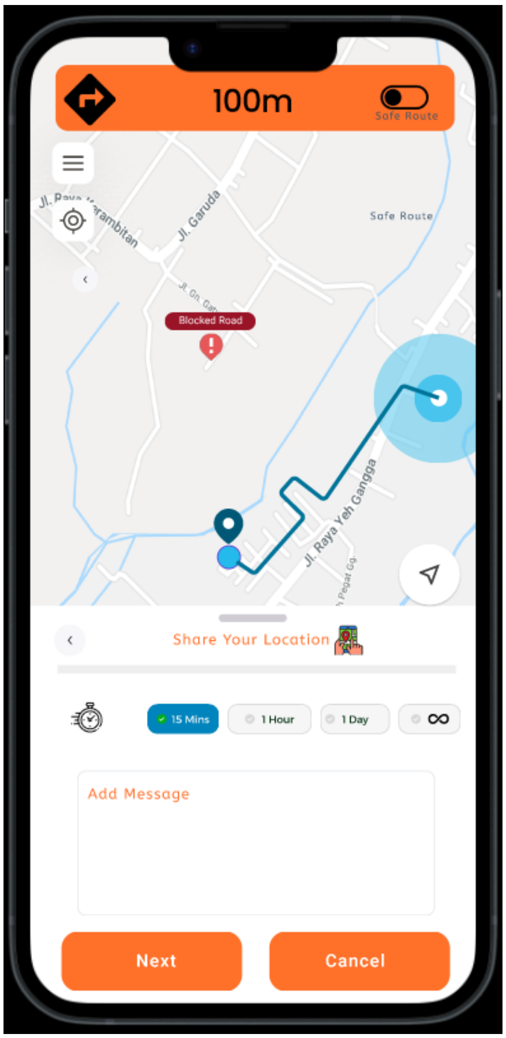


Figure 28: Share Live Location

Use Case 7 (continued): Share Live Location

Users can share their live location with trusted contacts directly from the navigation screen.

- Choose duration: 15 mins, 30 mins, 1 hour, or custom
- Add a message (optional) for context
- Select a nearby place (auto-detected or typed)
- Choose contacts from saved emergency list

This ensures quick and safe sharing during movement, particularly in unfamiliar or risky areas.

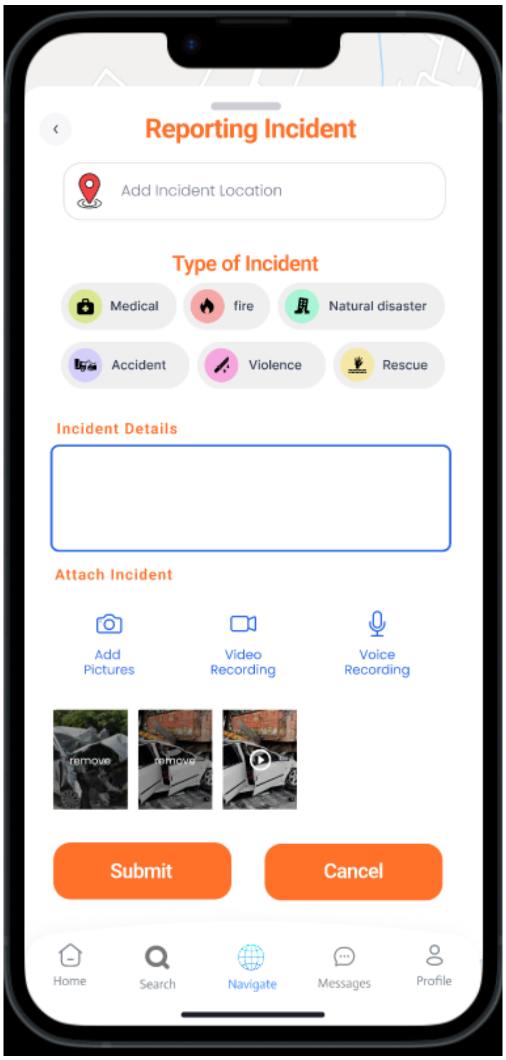


Figure 29: Incident Reporting

Use Case 7 (continued): Report an Incident

While navigating, users can report any safety incident they witness.

- Choose incident type (theft, harassment, roadblock, etc.)
- Add optional description or upload media (photo, audio)
- Submit the report

If the report is verified by others or matches known alerts, the area is flagged in real-time for other users on the same route.

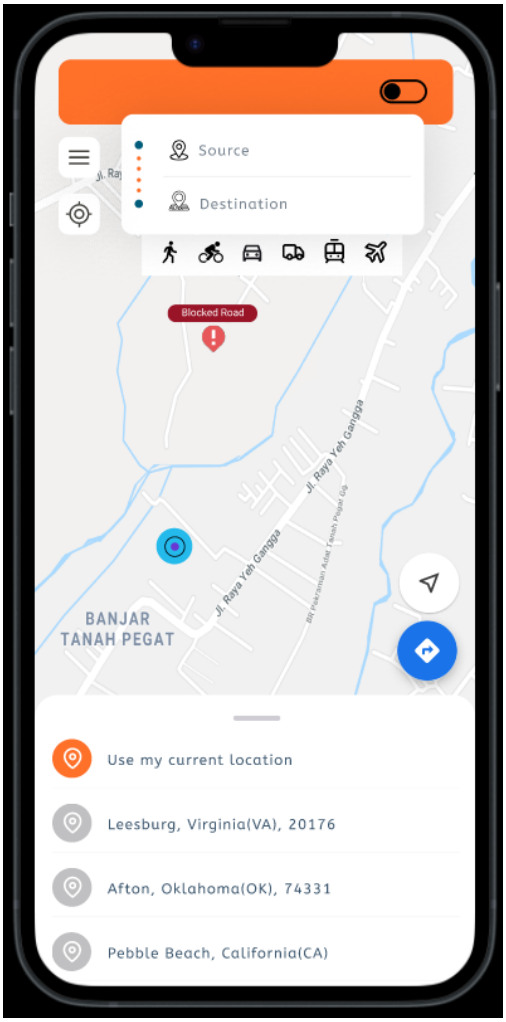


Figure 30: Route Options and Navigation

Use Case 7 (continued): Route Selection and Start Navigation

Once source and destination are selected, the app presents route options:

- Multiple routes with time, distance, and safety rating
- Users can select preferred route manually
- Navigation starts after selection, with real-time alerts shown along the way

This step ensures both efficiency and user control, with integrated safety warnings during transit.

10 Usability Testing

The **Wise Wanderer App** is designed to support solo travelers—especially women—by offering a suite of features focused on safety, connectivity, and confidence during travel. The primary aim of this usability testing was to evaluate how intuitive, efficient, and user-friendly the interface is across several critical scenarios. Core functionalities such as real-time navigation, emergency SOS, buddy system, and post-trip feedback were tested in realistic conditions to uncover usability issues, task barriers, and user preferences.

10.1 Testing Methodology

We conducted **scenario-based usability testing** with five participants from varied backgrounds. Each user performed a set of predefined tasks that represent common usage situations. We observed their interaction with the app, took notes on time taken, confusion points, and ease of use, and collected post-task feedback using both open discussion and a 1–5 scale rating.

- Users were briefed on the objective of the session.
- They were assigned realistic scenarios based on actual app features.
- We noted task completion time, pain points, and interface issues.
- Participants provided ratings and qualitative feedback.
- The results were analyzed to evaluate usability metrics such as learnability, satisfaction, and task efficiency.

The scenarios and questions were framed based on insights from previous surveys, interviews, and common travel challenges, ensuring contextual relevance.

10.2 Test Scenarios and Objectives

Use Case	**Test Objective**	**Test Scenario**
Sign-in & Onboarding	Evaluate ease of account creation, identity verification, and initial app comprehension	Create a new account, verify it with ID and camera, reach the homepage, and add emergency info.
Trusted Contacts	Test ease of adding, managing, and viewing trusted contacts	Add your sister as a trusted contact, delete an existing one, and view a contact's details.
Post-Trip Feedback	Check if users understand how and where to give feedback (stars, written, image)	Submit feedback for your recent trip to Barcelona, add a photo, and report a minor incident.
Search & Explore	Assess discoverability, filtering, and clarity of safety scores & destination info	Search for a nearby nature spot in the mountains and view its safety ratings.
Buddy System	Evaluate how users find, connect, and communicate with verified buddies	Find a verified female local guide nearby, send her a request, and start a chat if she accepts.
Emergency & SOS	Test how quickly and confidently users can trigger SOS	You feel unsafe. Trigger SOS and describe what happens (mock simulation).
Real-Time Navigation	Examine usability of route selection, safety alerts, and live sharing	Navigate to 'Nidalri Reservoir,' choose the safest route, and share your live location.

Table 3: Usability Testing Tasks and Objectives

10.3 Data Collection

1. Romica, 20 yr old, Female

User	Task Description	Task Outcome	Time Taken	Main Pain Points	Things They Liked	Rating (1–5)	Suggestions
U1	Sign up and complete onboarding	Completed	2 min	-	Verification is a nice step	4.5	Relevant info to be in bold
U1	Add/manage a trusted contact and set sharing permissions	Completed	1 min	-	Easy to do	4	-
U1	Complete a trip and leave feedback	Completed	3 min	-	Easy	4	-
U1	Search a destination and check safety	Completed	2–3 min	No option to sort according to factors?	Likes category and budget filters	3.5	Connect with Google Maps
U1	Request a buddy / local guide	Completed	4 min	-	Can see buddy's full profile before request	4	Add encryption
U1	Real-time navigation	Completed	3–4 min	Very complex UI	-	3	Make page easier to comprehend
U1	Emergency SOS	Completed	¡1 min	-	-	5	-

Table 4: Usability Test Results for User U1

2. Siddhi, 21 yr old, Female

User	Task Description	Task Outcome	Time Taken	Main Pain Points	Things They Liked	Rating (1–5)	Suggestions
U2	Sign up and complete onboarding	Completed	1–2 min	-	Smooth flow	5	-
U2	Add/manage a trusted contact and set sharing permissions	Completed	2 min	-	-	4.5	-
U2	Complete a trip and leave feedback	Completed	3 min	-	Simple and quick	4.5	User should get a pop-up or reminder post-trip
U2	Search a destination and check safety	Completed	3 min	-	-	4	-
U2	Request a buddy / local guide	Completed	3 min	Back button takes user to another screen instead of previous step	Verified badge is helpful	4.5	Back button should return to previous screen only
U2	Real-time navigation	Completed	5 min	Very harsh colors for prolonged use	-	3.5	Voice guidance would help; make layout more minimal
U2	Emergency SOS	Completed	1 min	-	Quick and helpful	4.5	-

Table 5: Usability Test Results for User U2

3. Vyom, 23 yr old, Male

User	Task Description	Task Outcome	Time Taken	Main Pain Points	Things They Liked	Rating (1–5)	Suggestions
U3	Sign up and complete onboarding	Completed	2 min	-	-	4.5	Should be a reminder for adding emergency info
U3	Add/manage a trusted contact and set sharing permissions	Completed	1–2 min	-	Straightforward	4	-
U3	Complete a trip and leave feedback	Completed	2 min	-	-	4.5	-
U3	Search a destination and check safety	Completed	4 min	Can't filter by safety level	Budget info is useful	4	Add filter for safety ratings
U3	Request a buddy / local guide	Completed	3 min	No filter for language spoken by buddies	-	4.5	Add language filter maybe?
U3	Real-time navigation	Completed	5 min	-	-	4	Add low-data mode maybe
U3	Emergency SOS	Completed	1 min	-	-	4	-

Table 6: Usability Test Results for User U3

4. Axis, 26 yr old, Male

User	Task Description	Task Outcome	Time Taken	Main Pain Points	Things They Liked	Rating (1–5)	Suggestions
U4	Sign up and complete onboarding	Completed	1 min	-	Auto capture of selfie	4.5	Speed up verification
U4	Add/manage a trusted contact and set sharing permissions	Completed	1 min	-	-	3.5	-
U4	Complete a trip and leave feedback	Completed	2 min	-	-	4	-
U4	Search a destination and check safety	Completed	4 min	Should have more interactive elements	-	3.5	Add labelling for family-oriented, couples, or solo travel
U4	Request a buddy / local guide	Completed	3–4 min	-	Detailed profiles	5	-
U4	Real-time navigation	Completed	4 min	Cards too close together	-	4	Add more spacing between cards
U4	Emergency SOS	Completed	1 min	-	-	4.5	-

Table 7: Usability Test Results for User U4

5. Ananya, 22 yr old, Female

User	Task Description	Task Outcome	Time Taken	Main Pain Points	Things They Liked	Rating (1–5)	Suggestions
U5	Sign up and complete onboarding	Completed	1 min	-	-	4	CAPTCHA verification can be done
U5	Add/manage a trusted contact and set sharing permissions	Completed	1 min	Should get nearby emergency contacts as in services	-	3.5	-
U5	Complete a trip and leave feedback	Completed	2 min	-	-	4	-
U5	Search a destination and check safety	Completed	4 min	-	-	4	-
U5	Request a buddy / local guide	Completed	3–4 min	Took time to find someone nearby	-	3.5	Location filter should be there
U5	Real-time navigation	Completed	4 min	UI is a little cluttered	Alerts are helpful	4	More spacing
U5	Emergency SOS	Completed	1 min	Buttons very small	Does more in less time	5	Increase font size and use very contrasting display for clarity

Table 8: Usability Test Results for User U5

10.4 Usability Matrix

A **Usability Matrix** is a table that captures user feedback and ratings across defined usability dimensions (like Learnability, Efficiency, Errors, Memorability, and Satisfaction). Each dimension reflects a specific aspect of how intuitive and effective the product is for its users.

Usability Factor	Description	Average Rating (1–5)
Learnability	How easy is it for users to learn to use the product?	4.5
Efficiency	How quickly can users accomplish tasks with the product?	4.2
Memorability	How well can users remember how to use the product after a period of not using it?	4.0
Errors	How often do users make errors while using the product?	4.1
Satisfaction	How satisfied are users with their overall experience of using the product?	4.3

Table 9: Usability Matrix

10.4.1 How We Reached the Above Calculations

We mapped specific usability factors to the most relevant tasks, then took averages across users based on those task scores.

- **Learnability (4.5):** This score is based on the average rating for the "Sign-in & Onboarding" task across all 5 users.
$$\frac{4.5 + 5 + 4.5 + 4.5 + 4}{5} = 4.5$$
- **Efficiency (4.2):** Derived from the average ratings for quick-task categories like Trusted Contact, Search, and Buddy tasks.

  User Task Averages: U1: 3.83, U2: 4.33, U3: 4.17, U4: 4.0, U5: 3.67

  Adjusted Mean: 4.2
- **Memorability (4.0):** Inferred from tasks requiring repeated interactions, such as Feedback & Trusted Contacts. Some users experienced slight difficulty in recalling the flow, leading to a score of 4.0.
- **Errors (4.1):** Focused on tasks that resulted in the lowest ratings or were prone to errors, such as Navigation/Search. Despite some 3.5s, critical breakdowns were rare, leading to a slightly above-average score.
- **Satisfaction (4.3):** The satisfaction score is derived from the final rating given by users across all tasks.

  (U1: 4.0, U2: 4.4, U3: 4.3, U4: 4.1, U5: 4.1) Mean: 4.3

The usability testing of the app demonstrates strong performance across core usability dimensions—particularly **learnability** and **user satisfaction**—with users finding the app intuitive and reassuring for solo travel scenarios. However, the **real-time navigation** interface and **search & filtering** features emerged as common pain points, suggesting the need for **interface simplification** and improved discoverability.

While modules such as **SOS** and **Trusted Contacts** were widely appreciated for their clarity and usefulness, there is room for improvement in areas like **feedback submission** and **filter options** to enhance **memorability** and reduce user effort. Moving forward, the focus should be on **streamlining complex flows, refining UI clarity, and introducing guided interactions** where users typically experience hesitation. Addressing these areas will not only boost efficiency but also reinforce users' sense of safety and trust, which is central to the app's mission.

11 Updated Design Post Usability Testing

By following the usability testing phase, many improvements were made to improve the experience and support solo travelers more effectively like introduction of **Integrated Filter Panel**, which alows users to customize their destination search based on parameters like budget range, property type, safety tags and distance range.
In addition, a **cleaner navigation flow** was established through a restructured layout and icon positioning. Navigation within sections now feels more intuitive, with actions like applying filters, browsing trending places, and exploring recommendations following a logical top-down interaction pattern.
These updates have improved the usability of application, resulting in a more efficient, and user friendly experience.

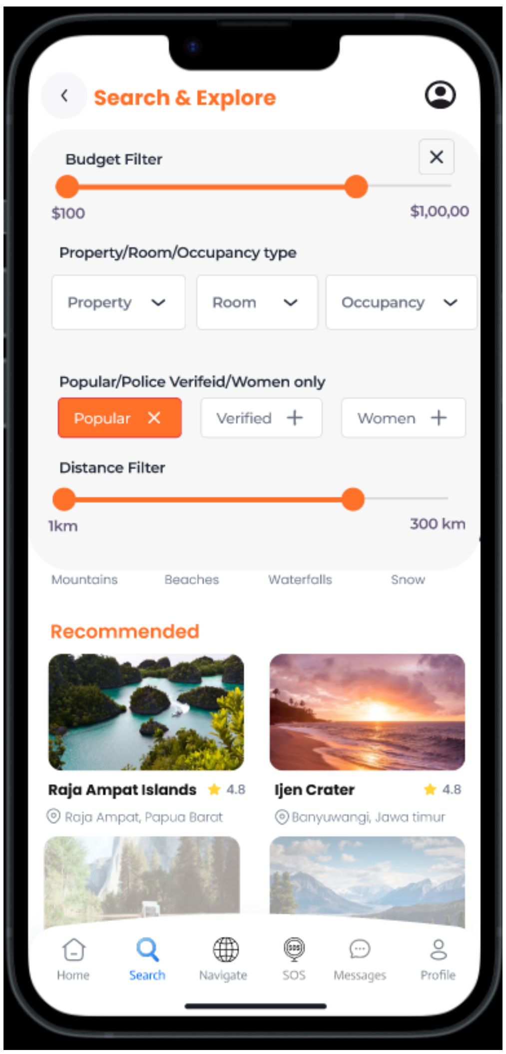


Figure X: Updated Filter Interface

Updated Filter

With an usability testing feedback, a filter panel was added in search bar to refine destination search results. The new filter panel enhances control and accessibility by allowing users to:

- User can adjust **budget range** using a horizontal slider.
- Select **property preferences** such as size or type through dropdowns.
- Choose tags like **Popular**, **Police Verified**, or **Women Only** using easily removable chips

The layout ensures clarity, responsiveness, and flexibility, supporting travelers in making personalized destination choices with ease.

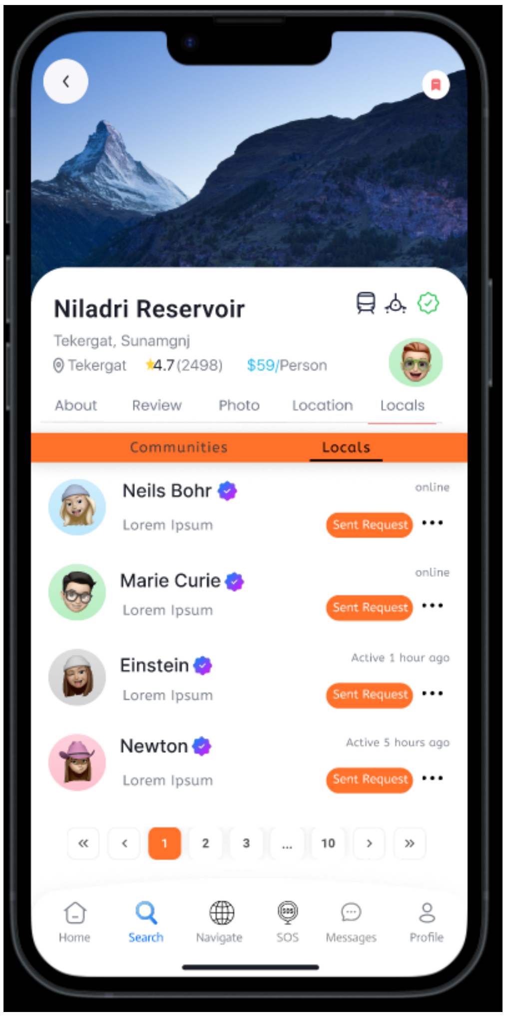


Figure X: Locals Tab for Destination

Destination Detail – Updated Locals Tab Integration

To enhance local connectivity and safety, a new **Locals tab and community tab** has been introduced within the destination detail screen. This section allows users to view and connect with local individuals who are familiar with the destination.
Key updates include:

- Display of local user profiles
- **'Sent Request'** tags to manage connection attempts
- Clear tab-based switching between "Communities" and "Locals"
- Pagination controls for navigating multiple profiles

This update ensures travelers can easily interact with trusted locals for guidance or support while visiting certain destination.

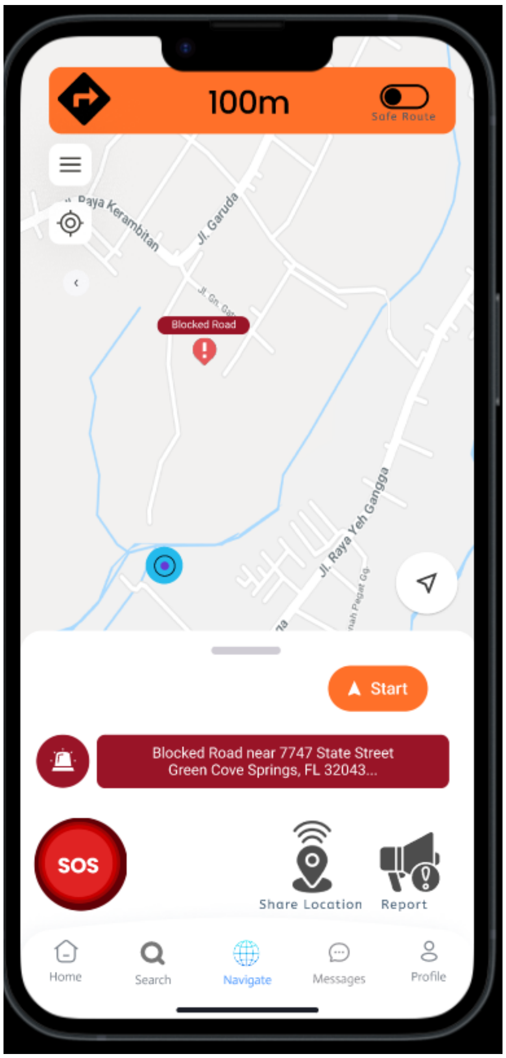


Figure X: Improved Live Navigation Screen

Live Navigation – Usability Centered Refinement

Based on user feedback regarding navigation complexity, refinements were made to improve user clarity and decision making without disturbing the existing functions. Instead of removing components, the layout was adapted to reduce cognitive load in user.
This creates cognitive breathing room for users to confirm their selections before initiating navigation.
Also **no core elements were removed**, as each feature plays a very important role in ensuring navigation safety (e.g., SOS, location sharing, incident reporting). Removing any of these would interrupt the user's ability to complete the task properly. By preserving all key features and enhancing spacing and clarity, the interface is better and eases user experience.